Better Jobs for Mothers, More Sons at Birth*

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Abstract

Vietnam's sex ratio at birth rose from 107 boys per 100 girls to 112 as trade liberalization expanded female employment, even though women's economic gains are usually expected to favor daughters. Standard accounts emphasize fertility decline, relative returns, and income gains, but do not distinguish maternal from paternal labor-market gains. We develop a model in which maternal wages, unlike paternal wages, raise the cost of children because mothers bear childcare time costs, lowering fertility, and a quantity-quality tradeoff can amplify this channel as fertility falls. Using industry-level tariff cuts from the 2001 US-Vietnam trade agreement, we assign exposure separately to each parent. Maternal exposure raises male birth probability and hours worked and lowers fertility. Paternal exposure raises fertility but does not affect sex selection. This joint pattern supports a maternal time-cost channel over income-only or relative-returns explanations.

JEL Classification: F13, J13, J16, J22

Keywords: Sex ratio at birth, Sex selection, Fertility, Female labor supply, Trade liberalization

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1. Introduction

Vietnam's sex ratio at birth rose from 107 boys per 100 girls in 1999 to 112 by 2015. Over the same period, female wage employment nearly doubled as a share of total female employment, fertility remained low, and women's university enrollment rose relative to men's (Figure 1). Similar combinations of rising sex ratios and expanding labor-market opportunities for women appear in other countries with son preference during periods of trade integration.

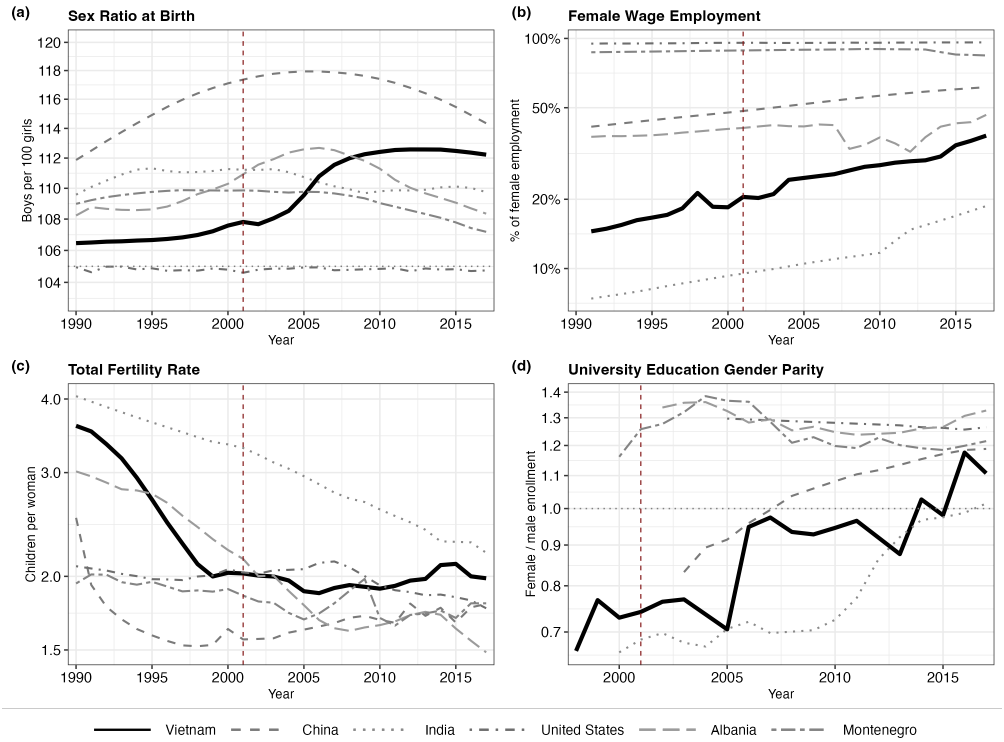
These patterns present a puzzle. An influential view is that economic gains to women often improve child outcomes and can benefit daughters in particular (Jayachandran and Voena, 2026), whether through grandmother pensions (Duflo, 2003) or higher female income in tea-growing regions (Qian, 2008). Yet sex ratios worsen precisely when women gain access to better jobs. The existing literature offers three explanations for rising sex ratios. First, fertility reductions combined with son preference mechanically raise the sex ratio, because families with fewer births are more likely to use sex selection to ensure at least one son (Jayachandran, 2017). Second, higher relative returns to female work raise the economic value of daughters, reducing sex selection (Rosenzweig and Schultz, 1982; Qian, 2008). Third, income gains favor sons, because sex selection is a normal good that wealthier families can better afford (Almond et al., 2019). None of these accounts distinguishes between maternal and paternal income sources. That omission matters because it leaves unexplained why better jobs for mothers coincide with rising sex ratios, falling fertility, and a closing education gender gap.[1]

In this paper we show that the source of household income matters for sex selection. Maternal and paternal labor market gains have opposite effects on fertility and different effects on sex composition. To understand why, we develop a model that incorporates the fertility-decline, relative-returns, and income mechanisms but adds one crucial asymmetry. Because mothers are the primary providers of childcare (Doepke et al., 2023), a maternal wage gain raises the time cost of children in a way a paternal wage gain does not. Maternal wages therefore create pressure to reduce fertility and to concentrate remaining births on sons, whereas paternal wages raise household resources without tightening the maternal time constraint.

The model then adds a quantity-quality tradeoff that amplifies this time-cost channel. As fertility falls, parents invest more in each remaining child. When the maternal wage rises, daughters' future labor market returns also rise, prompting heavier investment in

[1] Marriage-market channels through which relative wealth or dowry practices affect sex ratios (Borker et al., 2024) are not first-order in Vietnam, where dowry is uncommon.

Figure 1: Sex Ratio, Female Wage Employment, Fertility, and Education Gender Parity



Notes: Data from Chao et al. (2019) (panel a) and the World Bank World Development Indicators (panels b–d). Vietnam is the solid black line. Vertical dashed line marks 2001 (BTA). Horizontal dotted line in panel (a) marks the natural ratio (≈105). Panel (d) is the Gender Parity Index for university enrollment (female-to-male ratio). Log scale on all vertical axes.

daughters and increasing the total education cost per child. That higher per-child cost reinforces the pressure to reduce fertility and concentrate remaining births on sons. An increase in the paternal wage does not generate the same amplification. Higher paternal wages raise household income, but they also make sons more expensive to educate, offsetting rather than reinforcing the income effect.

This framework provides a mechanism for the regularity emphasized by Jayachandran (2017), namely that fertility decline accounts for a substantial share of rising sex ratios. The key testable contrast is simple. Maternal labor market gains should increase sex selection and reduce fertility, while paternal gains should raise fertility without affecting sex composition. That joint pattern distinguishes the maternal time-cost channel from income-only and relative-returns explanations.

We test these predictions using the 2001 US-Vietnam Bilateral Trade Agreement (BTA), which generated industry-specific tariff cuts that reached mothers and fathers differently

depending on where each parent worked. We combine the 1999 Vietnam Population Census with the Vietnam Household Living Standards Surveys (VHLSS, 2002–2010), which record each parent's industry and employment at the time of birth. Observing both parents contemporaneously is the key advantage of these data. By including maternal and paternal industry exposures simultaneously, we can separate shocks operating through the mother's job, the father's job, and the broader local economy.

The estimates match the model's discriminating pattern. At the median industry tariff cut, maternal exposure raises male birth probability by about 2 percentage points. The binary specification yields the same positive pattern throughout. Mothers in high-exposure industries are 15 percentage points more likely to hold a wage job, from a base of 25 percent, and work approximately 15 percent more hours per month. Annual birth probability falls by 0.6 percentage points, a 30 percent decline relative to the mean. Exposure through fathers' industries or local labor markets has no detectable effect on sex selection, though paternal exposure raises fertility by about 25 percent, consistent with the model's prediction that paternal income operates without tightening the maternal time constraint.

The sex selection literature documents apparently contradictory patterns (Sen, 1990; Das Gupta et al., 2003; Ebenstein, 2010; Guilmoto, 2009). In China, income gains raise sex selection (Almond et al., 2019), yet higher relative returns to female work reduce it (Qian, 2008). Our model reconciles these findings by showing that the same framework can generate both patterns depending on whether fertility can adjust. Under binding constraints like China's one-child policy, income shocks channel directly into sex selection, while high per-child investment amplifies the value of daughters when female returns rise. In less constrained settings like Vietnam, the maternal time-cost channel dominates.

A common empirical approach in the sex-selection literature uses geographic variation that assigns the same shock to all households in a location (Qian, 2008; Anukriti and Kumler, 2019; Chakraborty, 2015). By observing both parents' industries and employment at the time of birth, our data allow us to assign industry-level exposure separately to mothers and fathers and to test directly whether the source of income matters.

The fertility and human capital literature typically treats the quantity-quality tradeoff and sex selection as separate problems (Rosenzweig and Wolpin, 1980; Rosenzweig and Zhang, 2009; Jayachandran and Kuziemko, 2011; Anukriti et al., 2022). Our model connects them. The same time-cost pressure that reduces fertility and raises sex selection also implies heavier investment in fewer children, while rising maternal wages increase the return on daughters' education. Figure 1, panel (d), and the evidence in Mitra et al. (2026) are consistent with this implication.

The Vietnam BTA literature (McCaig, 2011; McCaig and Pavcnik, 2018; McCaig et al.,

2025) documents formalization, poverty reduction, and firm entry following the trade agreement but has not examined demographic consequences. We show that the same trade shock that improved women's labor market outcomes worsened prenatal gender imbalance.[2]

Policies that only raise household income are unlikely to break the link between female employment and sex selection. In our framework, the operative channel runs through maternal time costs rather than income alone.[3] Policies that directly reduce the maternal time constraint, such as expanding access to institutional childcare, therefore address the mechanism through which female employment growth worsens sex ratios.

2. Model

The maternal time-cost mechanism rests on a specific economic structure that we now formalize. The key assumption, standard in the fertility literature (Jones et al., 2011; Doepke et al., 2023), is that mothers bear a disproportionate share of childcare time costs. Because mothers' time is the binding input, the mother's wage carries information beyond its contribution to household income. It represents the opportunity cost of the time she devotes to childcare, tightening her time constraint in a way that a paternal wage gain or non-labor income does not.

2.1 Baseline Model

Consider a household with pre-existing son preference, rooted in patrilineal kinship norms where sons provide old-age support and continue the family line (Belanger, 2002). The expected number of boys is $B = bn$, where n is total fertility and $b \in [0, 1]$ is the probability that a birth is male, which the household can influence through sex-selective practices.[4] The household chooses consumption c, the mother's leisure ℓ, total fertility n, and the

[2] Huynh and Ku (2025) use a province-level shift-share to show that the BTA reduced female labor force participation among married and wealthier women. Drozdoff et al. (2025) show that foreign manufacturing jobs delay marriage and childbirth through migration. Both papers study different margins of female-oriented labor demand. We study parental industry exposure at conception and show that, among births that occur, higher maternal opportunity costs reallocate births toward sons.

[3] Anukriti (2018) finds that a cash-transfer program in Haryana lowered birth rates but worsened the sex ratio at birth, as households ensured at least one son before stopping. Our design goes further by separating maternal and paternal income, showing not only that income gains that do not tighten the maternal time constraint leave sex selection unchanged, but why.

[4] The variable b translates directly to the sex ratio at birth: Sex Ratio $= 100 \times b/(1-b)$. For example, $b = 0.51$ corresponds to 104 males per 100 females, while $b = 0.52$ corresponds to 108.

male birth probability b to maximize

$$\max_{c,\ell,n,b} U(c,\ell,n,B) \quad \text{where } B = bn, \tag{1}$$

with U increasing and strictly concave in each argument.[5]

Each child requires γ units of the mother's time, which competes with market work and leisure.[6] The model abstracts from endogenous paternal time allocation by setting $h_d = 1$, so the father contributes earnings w_d but bears no childcare time cost. The relevant margin for identifying the mechanism is maternal childcare time, and men's labor supply is largely unresponsive to fertility (Angrist and Evans, 1998). The mother allocates her unit time endowment among leisure, market work, and childcare:

$$\ell + h_m + \gamma n = 1, \tag{2}$$

where h_m is her labor hours. She earns wage w_m, and the household receives non-labor income I from land, transfers, and remittances. Substituting the time constraint, the budget constraint is

$$c + w_m \ell + w_m \gamma\, n + p_b\, b\, n = W, \tag{3}$$

where $W = I + w_m + w_d$ is full household income and p_b is the cost per expected boy from sex selection.[7]

The first-order conditions with respect to (c, ℓ, n, b) are derived in Appendix A.1, which also provides the baseline comparative statics:

$$\frac{U_\ell}{U_c} = w_m, \qquad \frac{U_n + U_B\, b}{U_c} = w_m \gamma + p_b b, \qquad \frac{U_B\, n}{U_c} = p_b\, n. \tag{4}$$

At the optimum the b-terms in the fertility condition cancel, leaving $U_n/U_c = w_m\gamma$. The sex-selection condition simplifies to $U_B/U_c = p_b$ after dividing by n. The effective shadow prices are therefore $(1, w_m, w_m\gamma, p_b)$ for the four goods (c, ℓ, n, B). The maternal wage enters two of the four prices. The paternal wage enters none, affecting demands

[5] U is twice continuously differentiable with $U_c, U_\ell, U_n, U_B > 0$ and the bordered Hessian conditions for a unique interior maximum satisfied.

[6] In our data, Vietnamese women spend over five hours per day on childcare compared to men's three hours (ActionAid-Vietnam, 2016). Formal childcare services remain limited outside major urban centers, and social norms assign childcare primarily to mothers.

[7] The income-side budget is $c + p_b bn \leq I + w_m(1 - \ell - \gamma n) + w_d$. Rearranging gives the full-income form. The parameter p_b encompasses the monetary costs of sex-determination technology (ultrasound fees, sex-selective abortion) and indirect costs such as health risks to the mother, psychological distress, and social stigma.

only through full income W.

The price ratio between fertility and boys,

$$\frac{U_n}{U_B} = \frac{w_m \gamma}{p_b}, \tag{5}$$

is the key object for comparative statics. When the maternal wage rises, $w_m \gamma$ increases and children become relatively more expensive compared to boys, pushing the household toward fewer children and higher sex selection.

Comparative statics. A rise in the maternal wage w_m affects sex selection through two reinforcing channels. The substitution effect operates through both price channels ($p_\ell = w_m$ and $p_n = w_m \gamma$), making leisure and children more expensive relative to sons. The income effect is weakly positive if the income elasticity of boys is at least as large as that of children. Both channels push b up, so $\partial b / \partial w_m > 0$. For fertility, the substitution effect is unambiguously negative, and when it dominates the income effect, $\partial n / \partial w_m < 0$. Maternal labor supply rises because a higher wage draws the mother into market work and lower fertility frees up childcare time.

The paternal wage w_d operates through income alone, since it enters no effective price. Under the same income-elasticity condition, the income effect on b is weakly positive. A pure income model would therefore predict that the paternal effect on b is at least as large as the maternal effect, because the father typically contributes a larger income share (Almond et al., 2019). The time-cost mechanism predicts the opposite, $\partial b / \partial w_m > \partial b / \partial w_d$, because the maternal channel adds a substitution effect the paternal channel lacks. The paternal wage raises fertility ($\partial n / \partial w_d > 0$) and mother's leisure ($\partial \ell / \partial w_d > 0$) through a pure income effect. Non-labor income I operates through the same channel, with a negligible effect on sex selection because it represents a small share of full income W.

The baseline model treats the cost of each child as pure time.[8] In practice, wages also determine how much parents invest in children's education. Figure 1 documents a closing education gender gap (panel d) alongside rising sex ratios (panel a), expanding female wage employment (panel b), and falling fertility (panel c) in Vietnam. The following extension connects these trends.

[8] The baseline model also abstracts from endogenous bargaining power and heterogeneous preferences across spouses. Appendix A.3 considers this extension.

2.2 Human Capital Investment

The intuition for this extension is as follows. When maternal wages rise, fertility falls and parents invest more in each remaining child. Because daughters' labor market returns also rise with the maternal wage, parents increase investment in daughters especially, raising total education costs per child. The higher per-child cost reinforces the time-cost pressure to reduce fertility further. At the same time, higher daughter returns make daughters more valuable, partially counteracting the incentive to select for sons. The formal analysis below shows that the cost-amplification effect operates through multiple reinforcing price channels while the daughter-value effect provides only a single offsetting force, so the net maternal effect on sex selection is likely positive. The asymmetry with the paternal side is equally important. When paternal wages rise, sons' returns increase and parents invest more in sons, but the resulting increase in son-specific education costs makes sex selection more expensive, not less. The paternal wage therefore creates an offsetting force rather than an amplifying one.

Parental wages serve a dual role. Beyond determining time costs, they signal the labor market returns to children's human capital. Children's labor market outcomes correlate with same-sex parental earnings, so maternal wages proxy for the return on daughters' education and paternal wages proxy for the return on sons' education. We formalize this by setting $r_g = w_m$ and $r_b = w_d$, where r_g and r_b denote the returns to girls' and boys' human capital.[9] This extension embeds our model in the quantity-quality framework of Becker and Lewis (1973), where the total cost of children's education scales with both the number of children and the investment per child.

Parents invest e_b in each son and e_g in each daughter, where e measures educational input (such as years of schooling). A human capital production function $h(e)$ converts educational input into effective units of human capital, with $h' > 0$, $h'' < 0$, and $h(0) = 0$. The market return r prices each efficiency unit, so a child's human capital value is

$$H_b = r_b \cdot h(e_b), \qquad H_g = r_g \cdot h(e_g). \tag{6}$$

Parents value children's human capital through $V(\bar{E})$ appended to the baseline utility, where $\bar{E} = b\, r_b\, h(e_b) + (1-b)\, r_g\, h(e_g)$ is the expected per-child human capital and V is

[9] The identification $r_g = w_m$ is most natural for individual-level variation, where a mother's own industry determines both her wage and the labor market signal for daughters' future earnings. At the province level, returns may differ from individual wages when local industry composition shifts daughters' future prospects independently of the mother's own wage. Appendix A.2.4 generalizes the model to allow province-level returns to differ from individual wages, and Section 7 tests this empirically using female-weighted provincial exposure P_p^w.

increasing and concave. The household now chooses $(c, \ell, n, b, e_b, e_g)$ jointly to maximize

$$U(c, \ell, n, B) + V(\bar{E}) \tag{7}$$

subject to the budget constraint

$$c + w_m \ell + n\left[w_m \gamma + p_e(b\, e_b + (1-b)\, e_g)\right] + p_b\, b\, n = W, \tag{8}$$

where p_e is the unit cost of educational input (tuition, school fees). The key feature of this budget constraint is that education costs enter as $n \times p_e \times (b\, e_b + (1-b)\, e_g)$, so quantity and quality are multiplied together. Adding one more child means paying for that child's entire education. Adding one more year of schooling means paying that cost for every child. Each margin raises the cost of the other, so parents trade off between the two.

Sex composition enters through the same interaction. Replacing a daughter with a son raises total education spending when $r_b > r_g$, because sons receive more investment. The budget tightens, so parents reduce schooling investment per child at the new optimum. The dependence runs in the other direction as well. When n falls, parents raise schooling investment per child, which in turn raises the cost of an additional child. This two-way dependence between quantity and quality is the tradeoff formalized by Becker and Lewis (1973).

The first-order conditions for e_b and e_g (Appendix A.2) equate the marginal return per dollar across genders:

$$r_b\, h'(e_b^*) = r_g\, h'(e_g^*). \tag{9}$$

Because h is concave ($h'' < 0$), the gender with higher market returns receives more investment. When $r_b > r_g$, $e_b^* > e_g^*$. When maternal wages rise relative to paternal wages, daughters' returns increase and the investment gap narrows, matching the closing education gender gap in Figure 1, panel (d).

Grouping terms by n and $B = bn$ decomposes the budget into a fertility component and a sex-selection component:

$$c + w_m \ell + n\left[w_m \gamma + p_e\, e_g\right] + B\left[p_b + p_e(e_b - e_g)\right] = W. \tag{10}$$

The fertility price $p_n = w_m \gamma + p_e\, e_g$ now includes both childcare time and daughter education costs. The sex-selection price $p_B = p_b + p_e(e_b - e_g)$ reflects the incremental cost of a son relative to a daughter. When $w_d > w_m$, sons receive more investment than daughters ($e_b^* > e_g^*$), so the education gap $p_e(e_b - e_g) > 0$ adds to the cost of sex selection beyond p_b.

Consider the effect of a rise in w_m on sex selection. The human capital extension introduces two channels that work in opposite directions. A cost channel operates through the effective prices. Higher $r_g = w_m$ raises daughters' education investment, which increases the fertility price p_n and likely narrows the education gap $e_b^* - e_g^*$ in p_B, amplifying the baseline time-cost effect. A value channel operates through $V(\bar{E})$. Higher daughters' earnings make them more valuable to the household, reducing the incentive to select for sons.

The cost channel pushes b up through multiple reinforcing price effects (the leisure price, the amplified fertility price, and the new sex-selection price), and the income effect reinforces the same direction, while only the value channel pushes b down. The net maternal effect on sex selection is likely positive. Falling fertility further amplifies the quantity-quality tradeoff, as lower n raises investment per child.

The extension also changes the paternal wage prediction. Higher w_d raises sons' returns ($r_b = w_d$), making sons more expensive to educate. This cost channel partially offsets the baseline income effect on b, pulling the net paternal effect toward zero. Nonlabor income I enters only through full income W without affecting education returns, so its effect on sex selection remains small. For fertility, neither w_d nor I enters p_n directly, so the income effect is the primary force and is positive. Higher income also raises desired child quality, increasing education spending per child and the effective fertility price p_n. As a result, the direct positive income effect on fertility is partially offset by a quality-substitution effect.

The model therefore generates a joint pattern consistent with the trends documented in Figure 1. Rising maternal wages reduce fertility, raise sex selection, and through the quantity-quality tradeoff, narrow the education gender gap.

These predictions apply to settings like Vietnam where fertility can adjust. Under binding fertility constraints, as under China's one-child policy, the model generates different predictions (Appendix A.2.5). When the fertility margin is constrained, the substitution pressure from a maternal wage increase that would otherwise reduce n channels into sex selection instead. Separately, income gains that have negligible effects on b in our setting may become positive predictors of sex selection, because the only way to respond to higher income is through the composition of births rather than their number. At the same time, high per-child investment (a consequence of the binding quantity constraint) amplifies the value channel. When female returns rise, daughters become substantially more valuable because each daughter receives heavy investment. The relative-returns channel, which is dominated by time costs in Vietnam, gains force when the fertility margin is shut down. The model therefore speaks to both settings. The time-cost predictions

we test apply to Vietnam. The constrained-fertility predictions are consistent with the findings of Almond et al. (2019) and Qian (2008) in China.

Testable predictions. Table 1 collects the comparative statics. We highlight four predictions that organize the empirical analysis.

Prediction 1 (Maternal time cost). A rise in maternal wages increases male birth probability ($\partial b/\partial w_m > 0$), raises maternal labor supply, and reduces fertility. All three responses reflect the same force, namely the higher opportunity cost of the mother's time.

Prediction 2 (Paternal income). A rise in paternal wages has at most weak effects on sex selection, because the education-cost channel offsets the income effect. The primary observable response is higher fertility ($\partial n/\partial w_d > 0$) through income.

Prediction 3 (Non-labor income). Non-labor income does not enter any effective price and represents a small share of full household income, so its effect on sex selection is quantitatively small. The effect on fertility is positive but likely small, because non-labor income is a limited share of full household income and part of the income effect is offset by higher desired child quality.

Prediction 4 (Discriminating pattern). The key test is the joint pattern across outcomes and exposure channels. Maternal exposure should increase male birth probability, raise maternal labor supply, and reduce fertility. Paternal exposure should mainly raise fertility. Provincial exposure should not reproduce the maternal combination once parent-specific channels are included. The income model of Almond et al. (2019) predicts that paternal gains produce sex-selection effects at least as large as maternal gains. The relative-returns mechanism of Qian (2008) predicts $\partial b/\partial w_m < 0$. Neither alternative generates the same joint pattern. Appendix A.3 considers a bargaining extension, and Section 7 tests all three alternatives.

3. Data and Measures

We implement the theoretical framework using Vietnam's 2001 Bilateral Trade Agreement (BTA) shock. This section describes how we construct exposure measures corresponding to the theoretical channels in Table 1, and then presents the household survey and census data used to observe parents' industries and birth outcomes. In the empirical design, maternal industry exposure M_i proxies for the maternal wage channel w_m, paternal industry exposure F_i proxies for the paternal wage channel w_d, and provincial shift-share exposure P_p proxies for the broad local income channel I.

Table 1: Model Predictions of How Income Sources Affect Household Outcomes

	Outcome		
	(1)	(2)	(3)
Increase in:	Sex Ratio	Maternal Labor	Fertility
	b (male share)	h_m	n
Mother's wage (w_m)	$+$	$+$	$-$
Father's wage (w_d)	$+/-$	$-$	$+$
Non-labor income (I)	0	$-$	$+$

Notes: $+$ positive, $-$ negative, $+/-$ ambiguous (opposing forces), 0 negligible. See Section 2 for discussion and Appendix A for formal derivations.

3.1 The 2001 BTA and Exposure Measures

Before the BTA went into effect on December 10, 2001, Vietnamese exports to the US faced high tariffs under the Column 2 schedule applied to non-market economies. The BTA granted Vietnam Most Favored Nation (MFN) status, dramatically reducing US tariffs on Vietnamese exports while having minimal impact on Vietnam's import tariffs.[10] Following the agreement, Vietnam's exports to the US surged from 0.7 billion dollars in 2000 to nearly 10 billion dollars in 2008.

We construct two complementary measures to capture how these differential tariff cuts transmit to fertility and sex selection through the labor market channels described in the model. The first captures direct exposure through parents' industries:

$$\tau_j = \log(1 + \text{Column } 2_j) - \log(1 + \text{MFN}_j), \tag{11}$$

where j denotes a parent's industry. A higher value of τ_j indicates a larger effective industry tariff cut. Because mothers and fathers tend to work in different industries, we construct separate measures for each parent to capture relative changes in labor market returns.[11] These industry-level measures proxy for the maternal and paternal wage chan-

[10] While Vietnam was obligated to reduce tariffs mainly on agricultural and food products, these reductions were minimal, ranging from 0.03 to 2.7 percentage points. In addition to tariffs, Vietnam agreed to eliminate various import quotas. Most of these quotas had, however, been eliminated by the end of 2002 (STAR-Vietnam, 2003).

[11] The correlation between maternal and paternal tariff exposure in our sample is consistently low, ranging from 0.24 to 0.36 across household survey years, described below, confirming that parents face different levels of exposure. The correlation between individual parental exposure and the provincial shift-share τ_p is even lower, remaining below 0.15 throughout the sample period. Appendix Figure C1 displays the full

nels (w_m and w_d) in Table 1. The measure captures genuine labor market improvements in the Vietnamese context because, using the same construction, McCaig and Pavcnik (2018) document large-scale reallocation from informal to formal employment following the BTA, and McCaig et al. (2025) show significant growth in both foreign and domestic firm entry in more exposed industries. We confirm this formalization channel directly in our own sample of parents at the time of fertility decisions below. Because female labor force participation was already approximately 70 percent before the BTA, the shock's labor market effects on employed mothers operated primarily through wages and hours rather than through new labor force entry.

The second measure is a province-level shift-share that captures broader local economic effects:

$$\tau_p = \sum_j s_{jp} \tau_j, \tag{12}$$

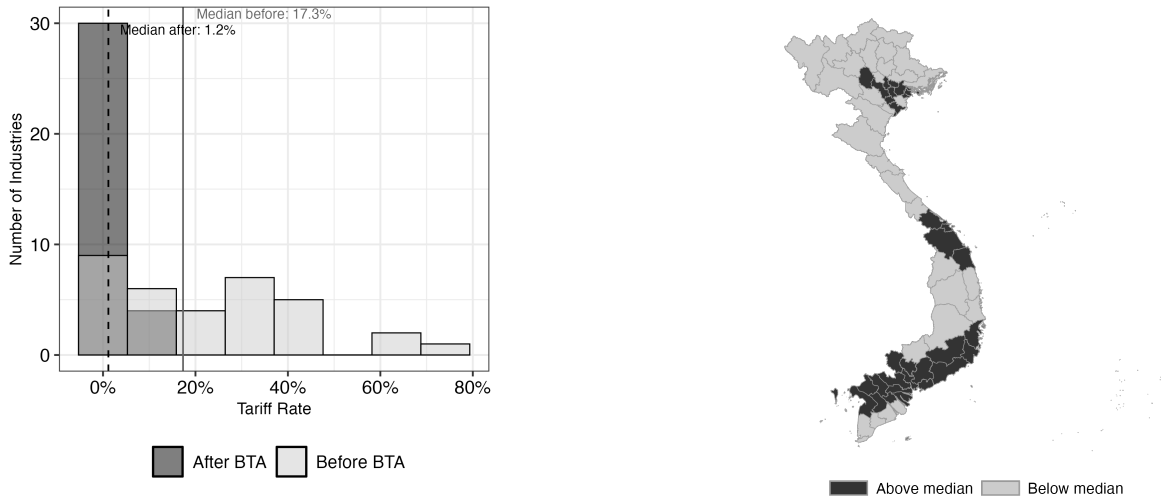
where s_{jp} is the share of province p's employment in tradeable industry j in the 1999 Census. Following Kovak (2013), employment shares are computed over all tradeable industries. This provincial measure captures broad local economic effects rather than any individual household's non-labor income. It proxies for the I channel in Table 1, but at the province level. Export expansion raises returns to land and agricultural assets, and income gains spread through intra-household transfers within extended families. Consistent with these channels, McCaig (2011) uses this province-level exposure measure to show that the BTA significantly reduced poverty rates across Vietnamese provinces, and McCaig et al. (2022) find that district-level variation in the same BTA shock led to improvements in children's human capital investment. The provincial measure τ_p therefore captures the reduced-form local effect of the trade shock, including gains that flow through household and community networks rather than individual labor markets alone.

Figure 2 illustrates the variation in both measures. Panel (a) shows the distribution of US tariff rates before and after the BTA across two-digit industries. The tariff reductions were large. The median Column 2 rate was 17.3 percentage points, falling to a median MFN rate of just 1.2 percentage points, a near-complete elimination of tariff protection. In log-difference terms, the industry tariff cut τ_j has a median of 0.168, reaching a maximum of 0.438 in the most exposed sector. A parent in the most exposed industry faced a tariff reduction roughly 2.6 times the median, implying substantially different incentives for occupational upgrading and wage growth across the treatment distribution.

Panel (b) maps the resulting province-level exposure. The measure varies by a factor of 3.6 across 61 provinces, from 0.067 in the least exposed to 0.239 in the most exposed

pairwise correlation structure by wave.

Figure 2: Variation in Trade Exposure Across Industries and Provinces



(a) Industry Tariff Changes

(b) Province-Level Exposure

Notes: Panel (a) shows US tariff rates on Vietnamese exports before (Column 2) and after (MFN) the BTA across 2-digit ISIC industries, from McCaig (2011). Panel (b) shades provinces by shift-share exposure $\tau_p = \sum_j s_{jp}\tau_j$, where s_{jp} is industry j's employment share in province p (1999 Census). Darker shading indicates above-median exposure.

(mean 0.089, median 0.081). The geographic pattern is economically intuitive. Above-median exposure is concentrated in southern provinces around Ho Chi Minh City and along the central and southern coast, areas with established export infrastructure and a higher pre-existing share of employment in manufacturing. Northern inland and mountainous provinces, where employment was concentrated in agriculture, show the lowest exposure.

For the main analysis, we convert these continuous measures into binary treatment indicators by splitting at their sample medians. We define $M_i = \mathbf{1}[\tau_{j(m)} > \tilde{\tau}]$, equal to one if the mother's industry tariff cut exceeds the sample median $\tilde{\tau}$, and zero otherwise, and define F_i analogously for the father. We define $P_p = \mathbf{1}[\tau_p > \tilde{\tau}_p]$, equal to one for provinces with above-median shift-share exposure.[12]

We prefer binary indicators for two reasons. First, standard two-way fixed effects regressions with a continuous treatment variable can assign difficult-to-interpret and potentially negative weights to heterogeneous treatment effects, and require a substantially

[12]Roughly 21 percent of parents work in industries where the Column 2 and MFN rates were both zero, yielding $\tau_j = 0$. These workers, concentrated in non-export service and agricultural sectors, fall in the low-treatment group.

stronger parallel trends assumption that rules out selection on potential outcome gains across treatment doses (Callaway et al., 2024). The binary specification identifies a well-defined high-versus-low-dose contrast under the standard parallel trends assumption, without this additional restriction. Second, the treatment effects have a straightforward interpretation as comparisons of high-exposure and low-exposure groups. Section 6 shows that continuous-treatment specifications produce consistent results.

3.2 Household Surveys and Census Data

To study how trade exposure affects sex ratio at birth, the ideal dataset would record each child's sex alongside both parents' industry, employment, and location around conception, allowing us to assign the trade shock separately to mothers, fathers, and provinces. The sex-selection literature typically relies on Demographic and Health Surveys (DHS) or population censuses, but each source introduces an identification failure that prevents separate channel identification.[13]

We therefore use the Vietnam Household Living Standards Surveys (VHLSS) (General Statistics Office of Vietnam, 2010), a biennial nationally representative survey spanning 2002 through 2010 that records industry, location, and employment for both parents at each survey wave. Because the VHLSS does not contain complete birth histories, we work with the flow of births in each wave, treating each cross-section as a sample of infants aged 0–1 observed alongside their parents. We supplement the VHLSS with the 1999 Population Census from IPUMS International (Ruggles et al., 2025), whose province-level employment shares serve as weights for the shift-share exposure measure and whose birth records extend the pre-treatment baseline.

The Census provides births from 1997–1999, contributing approximately 49,000 infant observations to the pre-treatment baseline. The VHLSS waves from 2002 through 2010 contribute repeated cross-sections of infants aged 0–1, yielding approximately 17,700 infant-parent observations across five waves. For each infant we observe the child's sex and the parents' contemporaneous industry codes and, where available, work hours and job types. Because the Census does not record work hours or wage employment status, only industry codes, it contributes to the sex selection analysis but not to the labor supply or fertility analysis. The fertility analysis uses the VHLSS roster reconstruction described below.

[13] DHS surveys provide birth histories, but they contain limited information on fathers and are conducted only intermittently. Population censuses are even more infrequent and come with a different problem: treatment must be assigned using industry and location at the census date rather than at conception. If workers switch industries or move after the shock, treatment is mismeasured and estimated effects are attenuated toward zero.

The pre- and post-treatment periods differ across outcomes. For sex selection and fertility, we define post-treatment as birthyear > 2001, so that Census births from 1997–1999 are entirely pre-treatment while children born in 2002 or later are post-treatment. For maternal labor supply, we define post-treatment as survey year > 2002, making the 2002 VHLSS the sole pre-treatment baseline for that outcome and waves from 2004 onward post-treatment.

A key measurement challenge in this setting is identifying parent-child relationships within extended family households, where multiple generations often live together and survey relationships are recorded relative to the household head rather than within nuclear families. We develop an algorithm that identifies both couple and parent-child relationships within extended households, validated against known parent-child links in the 2014 and 2016 VHLSS waves, achieving 93 percent accuracy for mothers and 91 percent for fathers. Appendix B details this procedure.

Table 2 presents summary statistics for Census 1999 and the five VHLSS waves. Mothers' baseline characteristics are comparable across sources, with average ages around 27–28 years and education ranging from 6.5 to 7.4 years across waves. The proportion of urban residents rises over time, consistent with Vietnam's urbanization trend. Male birth shares differ between the Census and the VHLSS. Census 1999 records 51.8 percent male births, corresponding to a sex ratio at birth of approximately 107 boys per 100 girls, while the 2002 VHLSS records 53.2 percent (SRB $\approx$ 113). The gap in sex ratios (approximately six SRB points, corresponding to a 1.4 percentage point difference in male birth shares) reflects the different time periods covered. Census births span 1997–1999, while the 2002 VHLSS captures births from 2000 through 2002, years in which the secular rise in sex selection was already underway, consistent with the upward shift documented in Figure 1. Across the 2004–2008 waves, the sex ratio remains elevated (SRB 110–112).

The model predicts that maternal and paternal wages have opposing effects on fertility (Table 1, column 3). Our main analysis conditions on the birth of an infant aged 0–1, so fertility, whether to have a birth at all, must be studied in a separate sample that includes woman-years without births. We reconstruct annual birth outcomes from the household roster. Because the roster records the birth years of all children currently living with the mother, we can infer in which years she had a birth. For each woman in a given survey year, we create one observation per year from when she turned 20 until the survey date, coding the outcome as one if a child in the roster has the corresponding birth year. The resulting sample contains approximately two million woman-year observations and supports mother fixed effects, which absorb time-invariant fertility preferences.

The main limitation of the retrospective measure is that children who have left the

Table 2: Summary Statistics for Census 1999 and VHLSS Survey Waves

	Census	VHLSS				
	1999	2002	2004	2006	2008	2010
Male birth	0.518	0.532	0.529	0.524	0.527	0.511
	(0.500)	(0.499)	(0.499)	(0.499)	(0.499)	(0.500)
Mother's age	27.00	28.10	28.39	27.88	28.09	28.24
	(5.05)	(5.19)	(5.29)	(5.27)	(5.13)	(5.04)
Mother's educ. (yrs)	6.46	6.72	7.17	7.42	8.01	8.22
	(3.43)	(3.60)	(3.71)	(3.69)	(3.71)	(3.66)
Ethnic minority	0.201	0.206	0.218	0.215	0.204	0.213
	(0.401)	(0.404)	(0.413)	(0.411)	(0.403)	(0.409)
Urban	0.131	0.175	0.211	0.231	0.266	0.270
	(0.337)	(0.380)	(0.408)	(0.421)	(0.442)	(0.444)
Birth order	1.93	2.07	2.02	1.90	1.88	1.89
	(0.81)	(0.80)	(0.80)	(0.80)	(0.78)	(0.76)
Mother wage earner	—	0.202	0.243	0.275	0.333	0.366
		(0.402)	(0.429)	(0.446)	(0.471)	(0.482)
Father wage earner	—	0.372	0.397	0.420	0.486	0.512
		(0.483)	(0.489)	(0.494)	(0.500)	(0.500)
N	48,897	5,934	3,159	3,034	3,164	2,374

Notes: Standard deviations in parentheses. Census 1999 covers 1997–1999 births. VHLSS waves include children aged 0–1. Dashes indicate variables unavailable in the Census.

household through marriage, migration, or death are absent from the roster, so earlier births are systematically missing for older women. The measure nonetheless provides a useful test of the fertility channel using the same treatment variation as the main analysis.

While migration across provinces does not contaminate the industry-level variation we exploit, workers may nonetheless switch industries in response to the trade shock, so that their current industry in the cross-section reflects post-shock sorting rather than pre-shock exposure. To address this selection problem, we exploit the VHLSS's rotating panel feature, which tracks a subset of households across three consecutive waves (2002–2004–2006). From these households we draw women aged 20–40 as our analysis sample. By fixing each parent's industry affiliation at the 2002 wave, the earliest available survey

and effectively a pre-BTA baseline given the short window since December 2001 implementation, we isolate trade exposure from compositional changes driven by post-shock worker reallocation. The panel structure also allows us to control for time-invariant unobservables, including fertility preferences and son preference, through individual fixed effects.

Appendix Table C1 compares the 2002-wave rotating-panel sub-sample ($N = 879$) to the main cross-section on baseline demographics. The balance table describes the 2002 baseline of the rotating panel, which is a subset of the 1,723 mothers used in the panel regressions across all three waves. Mothers' age, education, minority status, urban share, and province-level tariff exposure are closely matched across the two samples, suggesting that selection into the rotating panel is unlikely to explain differences between panel and cross-sectional estimates.

Given the short panel length, we rely on the cross-sectional variation for our main analysis and use the panel estimates as a robustness check.

4. Empirical Strategy

Testing the comparative statics in Table 1 requires an empirical design that separately identifies changes in maternal wages, paternal wages, and household non-labor income from a single trade shock. We embed the binary treatment indicators M_i, F_i, and P_p defined in Section 3.1 in a difference-in-differences framework that exploits the BTA's uneven tariff cuts across industries and provinces, comparing outcomes before and after the December 2001 implementation. Estimating all three channels jointly allows each coefficient to be identified conditional on the others, so that the resulting pattern across outcomes can discriminate among the competing mechanisms.

Following Section 3.2, we set $\text{Post}_t = \mathbf{1}[\text{birthyear} > 2001]$ for sex selection and fertility, and $\text{Post}_t = \mathbf{1}[\text{survey year} > 2002]$ for labor market outcomes. Because the BTA took effect in December 2001 and gestation takes approximately nine months, births in 2001 and early 2002 largely reflect pre-agreement conceptions, so the pre-treatment baseline extends into the formally post-treatment window.

Our baseline specification for sex selection estimates a linear probability model where the unit of observation is a birth:

$$y_{ipt} = \beta_1(M_i \times \text{Post}_t) + \beta_2(F_i \times \text{Post}_t) + \beta_3(P_p \times \text{Post}_t) + \alpha_t + \delta_p + \gamma_{j(m)} + \gamma_{j(d)} + X'_{ipt}\phi + \varepsilon_{ipt}. \tag{13}$$

Here y_{ipt} is an indicator equal to one if child i born in province p in year t is male. The

treatment indicators M_i, F_i, and P_p are assigned through the mother's industry, the father's industry, and the province of birth, respectively. The specification includes birth-year fixed effects α_t, province fixed effects δ_p, and ISIC industry fixed effects for both the mother $\gamma_{j(m)}$ and the father $\gamma_{j(d)}$. The control vector X'_{ipt} includes birth parity, the mother's age, education, urban residence, and minority status, world-average tariff controls at the parents' 2-digit ISIC industries matched by birth year, and interactions of pre-BTA provincial employment shares in agriculture and manufacturing with year fixed effects, which absorb differential trends arising from preexisting sectoral composition.[14] Standard errors are clustered at the province level.

For maternal labor supply, the unit of observation shifts to a mother-survey wave. We estimate by OLS:

$$y_{ms} = \beta_1(M_m \times \text{Post}_s) + \beta_2(F_m \times \text{Post}_s) + \beta_3(P_p \times \text{Post}_s) + \alpha_s + \delta_p + \gamma_{j(m)} + \gamma_{j(d)} + X'_{ms}\phi + \varepsilon_{ms}, \tag{14}$$

where y_{ms} is a labor market outcome for mother m observed in survey wave s, with $\text{Post}_s = \mathbf{1}[\text{survey year} > 2002]$. The fixed effects and treatment indicators mirror the sex-selection specification, except that world-average tariff controls are matched by survey year rather than birth year, and birth parity is excluded from the control vector because it is potentially endogenous to labor market conditions. All remaining controls, fixed effects, and clustering are as in Equation (13). The world-average tariff controls in both specifications absorb broader industry-level tariff movements during the study period, including contemporaneous multilateral liberalization around Vietnam's WTO accession in January 2007. The coefficients β_1, β_2, and β_3 capture the post-BTA differential between high- and low-exposure groups, net of all fixed effects listed above.

For fertility, the specification replaces province and industry fixed effects with mother fixed effects. The retrospective birth panel described in Section 3.2 yields approximately two million woman-year observations, with the outcome AnyBirth_{mt} equal to one if mother m had a birth in year t:

$$\text{AnyBirth}_{mt} = \beta_1(M_m \times \text{Post}_t) + \beta_2(F_m \times \text{Post}_t) + \beta_3(P_p \times \text{Post}_t) + \alpha_t + \mu_m + X'_{mt}\phi + \varepsilon_{mt}. \tag{15}$$

Mother fixed effects μ_m absorb all time-invariant heterogeneity, including underlying son preference, baseline fertility ideals, and permanent differences in industry and location. The control vector X'_{mt} includes the mother's age at birth and interactions of pre-BTA

[14] All specifications use 61 province clusters. Section 6 reports robustness to clustering at the industry level and to two-way clustering by province and industry.

provincial employment shares in agriculture and manufacturing with year fixed effects. Identification comes from within-mother variation in birth propensity before and after the BTA, comparing mothers with different levels of exposure.

Two threats to the cross-sectional estimates motivate the panel specification. The primary concern is industry switching. The cross-section records each parent's current industry rather than pre-shock affiliation, so workers who move into high-tariff-cut industries after the BTA can contaminate M_i and F_i with post-shock selection. Migration across provinces is a secondary concern. If the trade shock draws households with stronger son preference into high-exposure provinces, the composition of the high-P_p group shifts in a way that mechanically inflates β_3.

To address these concerns, we exploit the VHLSS rotating panel described in Section 3.2 and lock each parent's treatment status to her pre-BTA industry affiliation, following McCaig and Pavcnik (2018):

$$y_{ipt} = \beta_1(M_i \times \text{Post}_t) + \beta_2(F_i \times \text{Post}_t) + \beta_3(P_p \times \text{Post}_t) + \alpha_t + \mu_m + X'_{ipt}\phi + \varepsilon_{ipt}. \quad (16)$$

The unit of observation remains a birth i, as in Equation (13), but mother fixed effects μ_m replace province and industry fixed effects, absorbing all time-invariant maternal heterogeneity. Treatment indicators M_i and F_i are assigned based on each parent's industry reported in the 2002 VHLSS wave, the earliest available survey. Because the 2002 wave was fielded within months of the December 2001 BTA implementation and the VHLSS industry question refers to the respondent's current or most recent occupation, the industry recorded in 2002 effectively captures pre-BTA affiliation before meaningful occupational reallocation could have occurred. Fixing exposure at this initial wave ensures that post-BTA industry switching cannot alter assigned treatment, and the design mirrors an intent-to-treat experiment.

The panel sample consists of women aged 20–40 observed across the 2002, 2004, and 2006 VHLSS waves. For sex selection, we restrict to women who gave birth both before and after the BTA, so that within-woman comparisons can identify the treatment effect. If endogenous sorting were inflating the cross-sectional estimates, fixing treatment to pre-shock industries should attenuate the panel coefficients relative to the cross-sectional results.

The key identifying assumption is that, absent the BTA, outcomes would have evolved similarly across high- and low-exposure groups. Parallel trends is untestable in the post-reform period, but we can assess it directly in the pre-reform data. The 1999 Census births (1997–1999) and VHLSS births in 2000–2001 provide pre-treatment observations spanning

multiple years before the shock. The event study in Section 5 shows no differential trends between high- and low-exposure groups across these years, supporting the assumption. For the panel specification, we additionally assume that women's 2001 industry choices were not made in anticipation of the BTA's differential tariff effects across sectors.

The coefficients β_1, β_2, and β_3 map directly to the comparative statics in Table 1. The maternal time-cost mechanism predicts $\beta_1 > 0$ for sex selection and maternal labor supply and $\beta_1 < 0$ for fertility, because a rise in the mother's wage tightens her time constraint and intensifies the tradeoff between work and childcare. Paternal exposure (β_2) relaxes the budget constraint without altering the mother's opportunity cost of childrearing, so its effect on sex selection should be weaker than the maternal effect, with its main force appearing on fertility. Provincial income (β_3) does not enter any effective price and represents a small share of full household income, so its effect on sex selection should be smaller still. As Table 1 summarizes, the maternal time-cost mechanism is the only account that predicts positive sex selection, higher maternal labor supply, and lower fertility for the maternal channel, while paternal exposure mainly raises fertility and provincial income adds little.

5. Main Results

5.1 Effects on Sex Selection

Prediction 1 states that maternal exposure increases male birth probability. Table 3 tests this prediction alongside Predictions 2 and 3, reporting estimates from progressively richer specifications of Equation (13), where the dependent variable is an indicator equal to one if the child is male.

Columns (1) and (2) introduce the maternal and paternal exposure channels individually, while columns (3) and (4) combine them, with column (4) adding the full control set. Columns (5) through (7) introduce and combine the province-level channel. When entered alone in column (1), maternal industry exposure ($M_i \times \text{Post}_t$) raises male birth probability by 6.5 percentage points (s.e. 0.025), significant at the 5 percent level. Against a dependent variable mean of 0.52, this corresponds to a 13 percent relative increase in the probability that a birth is male.

For a mother whose industry received the sample median tariff cut of approximately 7 percentage points, the continuous treatment specification in Appendix Table C6 implies a 1.5 to 2.7 percentage point increase in male birth probability, depending on the control set. We treat this estimate as the magnitude benchmark because the binary specification pools heterogeneous doses into two groups. The raw means tell a similar story. The high-

Table 3: Effects of BTA Trade Exposure on Male Birth Probability

	MaleBirth						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$M_i \times$ Post	0.065**		0.058**	0.082**		0.057**	0.080**
	(0.025)		(0.025)	(0.032)		(0.024)	(0.032)
$F_i \times$ Post		0.040	0.020	0.034		0.020	0.032
		(0.031)	(0.031)	(0.030)		(0.031)	(0.030)
$P_p \times$ Post					0.013	0.003	0.015
					(0.013)	(0.012)	(0.016)
R^2	0.015	0.015	0.015	0.016	0.014	0.015	0.016
Observations	63,894	63,894	63,894	63,894	63,894	63,894	63,894
Dep. Var. Mean	0.518	0.518	0.518	0.518	0.518	0.518	0.518
Controls				✓			✓

Notes: Dep. var. is MaleBirth. M_i, F_i, P_p = 1 if maternal, paternal, or provincial tariff cut exceeds the sample median. All columns include birth-year, province, and parents' 2-digit ISIC FE. Controls (cols. 4, 7) add mother's age, education, urban, minority status, birth parity, world-average tariff controls at parents' industries, and pre-BTA provincial employment shares $\times$ survey-year. Province-clustered SE in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

exposure group's male birth share rose from 51.4 to 53.3 percent after the BTA, while the low-exposure group rose from 52.1 to 52.5 percent, a difference-in-differences of 1.5 percentage points. At the lower end of the continuous-treatment range, this corresponds to an increase in the sex ratio at birth from about 108 to 114 boys per 100 girls, close to the aggregate movement shown in Figure 1.

The father's industry exposure ($F_i \times \text{Post}_t$) enters with a positive but imprecisely estimated coefficient of 0.04 in column (2). In the combined specifications of columns (3) through (7), the paternal coefficient falls to 0.02 and remains statistically indistinguishable from zero, consistent with the model prediction that paternal wages have at most weak effects on sex selection, since they raise household income without tightening the maternal time constraint. Across these same specifications, the maternal coefficient moves from 0.065 in column (1) to 0.058 when the father is added and increases modestly to 0.082 with the full control set in column (4). The full control set includes the world-average tariff controls described in Section 4.

Province-level exposure ($P_p \times \text{Post}_t$) is near zero and insignificant throughout, en-

tering at 0.013 in column (5) and 0.018 in column (7). The model predicts this result. Non-labor income does not enter any effective price and constitutes a small share of full household income, so its effect on sex selection should be the weakest of the three channels. Section 7 examines whether these cross-channel patterns, taken together with the fertility and labor supply results, can distinguish the time-cost mechanism from alternative explanations.

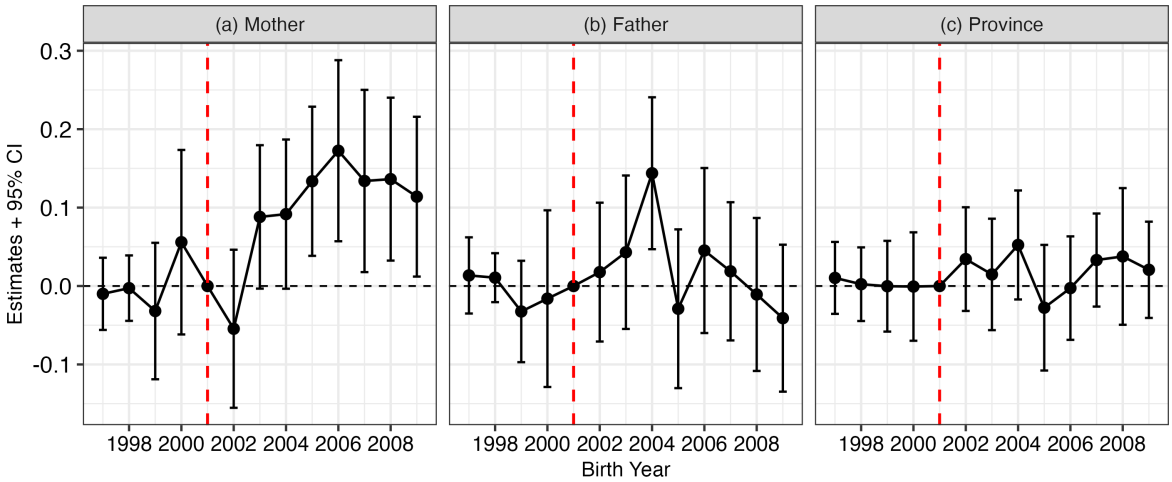
Column (7) presents the fully controlled specification with all three exposure channels entering simultaneously. The maternal coefficient ranges from 0.06 without controls to 0.08 with the full control set (columns 4 and 7, $p < 0.05$). To place the magnitude in context, Anukriti (2018) finds that first births become 0.5 to 1.2 percentage points more likely to be male after the Devi Rupak fertility incentive in India, and Almond et al. (2019) estimate a cumulative increase of approximately 2.8 percentage points over four years for second births following a firstborn girl in China. Our continuous-treatment estimate of 1.5 to 2.7 percentage points (Appendix Table C6) exceeds the upper bound of the range documented by Anukriti (2018) and falls below the benchmark from Almond et al. (2019), consistent with the fact that the latter conditions on the subgroup facing the strongest son preference pressure while our estimate pools across all birth orders and sibling compositions. The heterogeneity analysis below explores whether the maternal exposure effect varies across subgroups.

Figure 3 traces these effects over time, plotting event study estimates from the full specification where all three exposure channels enter simultaneously with the complete set of controls. Appendix Figure C2 presents individual event studies for each channel estimated separately, confirming that the patterns are not artifacts of joint estimation.

Panel (a) displays the maternal exposure coefficients. Pre-BTA estimates for the 1997–1999 Census cohorts and the 2000–2001 VHLSS births cluster tightly around zero, confirming that high- and low-exposure groups followed parallel trends well before the shock.[15] The coefficients remain indistinguishable from zero in 2002 and 2003. The BTA took effect in December 2001, so most births recorded in 2002 reflect conceptions that predate the agreement, and the 2002 VHLSS recall period straddles the policy change. The null coefficients through 2003 therefore extend the parallel trends validation into the post-announcement period. From 2004 onward, the maternal coefficients become positive and

[15] A reverse causality concern could arise because the design is cross-sectional. The mother's industry is measured contemporaneously rather than before the birth, so a mother who had a son could in principle sort into a high-treatment industry afterward. Two pieces of evidence rule this out. First, the flat pre-BTA coefficients show no pre-existing differential trend between high- and low-exposure groups, ruling out pre-existing compositional differences correlated with treatment. Second, if having a son led parents to work more and sort into higher-wage industries, the same mechanism should produce a positive paternal coefficient, but panel (b) shows no such effect.

Figure 3: Event Study Estimates of BTA Exposure on Male Birth Probability



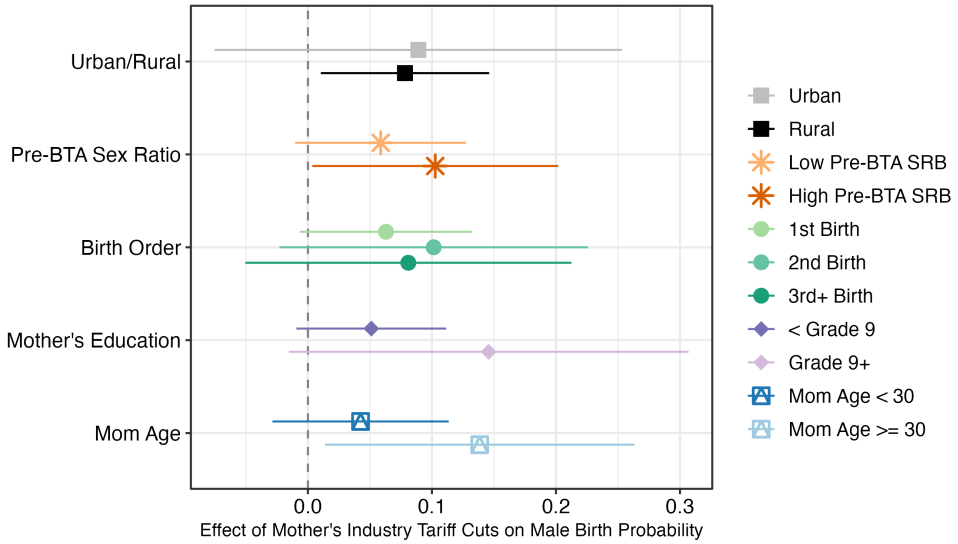
Notes: Coefficients from a single regression of Equation (13) with all three exposure channels and full controls. Panel (a): maternal (M_i), panel (b): paternal (F_i), panel (c): provincial (P_p). Binary indicators split at the sample median. Omitted year is 2001. Error bars are 95% CIs, province-clustered.

remain elevated. The two-to-three year delay between the BTA and the onset of effects is consistent with the time-cost channel. Although US tariff cuts took effect immediately, firms needed time to expand hiring in treated industries and women needed to observe the resulting wage improvement before adjusting conception timing. The nine-month gestation lag then pushes affected births into the 2004 wave and beyond.

Panels (b) and (c) display paternal and province-level exposure. Neither channel exhibits a consistent post-BTA pattern. Paternal industry exposure fluctuates around zero in most years, with the exception of an isolated 2004 spike that is a sampling artifact.[16] Province-level exposure is statistically insignificant throughout the study period. Pre-trend coefficients are flat for both channels, ruling out the concern that differential pre-existing trends could explain the maternal results through correlated shocks. The contrast across the three panels is the central identification result. The maternal time-cost mechanism predicts $\beta_1 > 0$ with β_2 and β_3 small for sex selection (Table 1, column 1). The event study confirms that the positive pattern is unique to maternal exposure and emerges only after the BTA could plausibly have affected labor market conditions. Because the main

[16]The high-father-treatment group for 2004 births comprises 198 observations spread across 146 province–industry cells, with 97 percent of cells containing at most 3 observations. The all-male draws in these tiny cells produce the spike. Adjacent years with similar cell structures do not show the same pattern, and the individual event study for the paternal channel confirms the spike is not an artifact of joint estimation.

Figure 4: Heterogeneous Effects of Maternal Industry Exposure on Male Birth Probability



Notes: Each point is the coefficient on $M_i \times \text{Post}_t$ from Equation (13) estimated within the indicated subgroup, with 95% CIs. Pre-BTA SRB split at median male birth share among 1997–1998 Census births. All regressions include full controls and fixed effects.

difference-in-differences specification defines Post as birth year after 2001, the pooled estimate includes 2002–2003 births whose conceptions largely predate the agreement. The null event-study coefficients for these transition cohorts imply that the reported DiD estimate is a conservative average that includes cohorts not yet affected by the BTA.

Heterogeneity. We examine how the maternal exposure effect varies across subgroups by estimating Equation (13) separately within each group. Figure 4 displays the coefficient on $M_i \times \text{Post}_t$ and its 95 percent confidence interval for each subgroup.

The maternal exposure effect is positive in both rural and urban areas, with point estimates of 0.078 and 0.089, respectively. The rural estimate is statistically significant ($p < 0.05$) while the urban estimate is imprecise, reflecting the smaller urban subsample. The absence of a strong urban-rural gradient is informative. If the mechanism operated primarily through differential access to sex-selective technology, which is more concentrated in cities, we would expect a larger urban effect. The similarity instead points to a time-cost channel operating through industry-level labor demand rather than through geographic access to medical services.

We split provinces by their pre-BTA sex ratio, using the median male birth share among 1997–1998 Census births as the cutoff (Figure C3 maps the provincial variation).

Provinces with higher pre-existing sex ratios show a larger maternal exposure effect (0.103, $p < 0.05$) than provinces with lower pre-BTA sex ratios (0.059, not significant), though the confidence intervals overlap. The directional pattern is consistent with the model's prediction. The pre-BTA sex ratio proxies for the intensity of pre-existing son preference across provinces. Where son preference is stronger, the marginal value of sex selection is higher, so a given increase in the maternal opportunity cost of childcare generates a larger response.

Birth order shows a positive gradient, with the coefficient on maternal industry exposure rising from 0.063 at first parity to 0.101 at second parity and 0.081 at third and higher parities.[17] None of the parity-specific estimates is statistically significant at conventional levels, but the pattern is directionally consistent with the time-cost mechanism. Higher-parity parents have fewer remaining births over which to fulfill son preference, so the marginal incentive to sex select at each birth is larger.

Mothers with higher education (ten or more years of schooling) show a larger point estimate (0.146) than those with less schooling (0.051), with the high-education estimate marginally significant ($p < 0.10$) and about 2.9 times as large. More educated mothers typically command higher wages in the labor market, so a given improvement in industry conditions implies a larger increase in their opportunity cost of time and a stronger incentive to economize on time-intensive childcare through sex selection. Neither estimate reaches the 5 percent threshold, so the education gradient should be read as suggestive rather than definitive.

Maternal age at birth generates the sharpest heterogeneity. Mothers aged 30 and above show a large and significant effect (0.139, $p < 0.05$), while mothers under 30 show no discernible response (0.042, not significant). Older mothers are more likely to be at higher parities without yet having a son, have accumulated work experience and face a higher opportunity cost of childcare, and confront a shorter remaining fertility horizon. Each of these factors reinforces the incentive to sex select and the pressure to produce a son before the fertility window closes.

The overall pattern is broadly consistent with the time-cost mechanism. Significant effects emerge primarily in subgroups where pre-existing son preference and the opportunity cost of childcare are jointly highest, namely provinces with high pre-BTA sex ratios and older mothers facing a narrowing fertility horizon. Where neither condition is particularly acute, the estimates are imprecise and no clear concentration is detectable.

[17] Births of order 3 or higher are uncommon in our data. Vietnam's total fertility rate had already fallen to 2.01 by 2000.

Table 4: Effects of BTA Exposures on Parental Labor Market Outcomes

	MomWageEmp		log(MomWorkHrs)		DadWageEmp	
	(1)	(2)	(3)	(4)	(5)	(6)
$M_i \times$ Post	0.149***	0.149***	0.159***	0.155***	-0.015	-0.014
	(0.027)	(0.027)	(0.051)	(0.051)	(0.029)	(0.029)
$F_i \times$ Post	0.018	0.018	0.000	-0.005	0.147***	0.148***
	(0.019)	(0.018)	(0.036)	(0.037)	(0.025)	(0.024)
$P_p \times$ Post		-0.003		0.054		-0.010
		(0.024)		(0.045)		(0.038)
R^2	0.604	0.604	0.254	0.254	0.530	0.530
Observations	17,389	17,389	17,198	17,198	17,389	17,389
Dep. Var. Mean	0.247	0.247	4.66	4.66	0.406	0.406
Controls	✓	✓	✓	✓	✓	✓

Notes: Dependent variables are maternal wage employment (cols. 1–2), log monthly hours (cols. 3–4), and paternal wage employment (cols. 5–6). Treatment indicators and fixed effects as in Table 3, excluding birth parity. Controls include world-average tariff controls at parents' industries (survey-year matched). Province-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

5.2 Effects on Parental Labor Market Outcomes

Prediction 1 requires that the BTA actually improved mothers' labor market conditions. If the industry-level tariff cuts did not translate into better jobs for mothers, the sex selection results would lack an economic channel. Table 4 tests this premise by estimating the same specification on parental labor market outcomes. Columns (1)–(2) report effects on maternal wage employment, columns (3)–(4) on maternal log monthly hours, and columns (5)–(6) on paternal wage employment.

Column (1) shows that mothers in high-tariff-cut industries are 15 percentage points more likely to hold a wage job after the BTA ($p < 0.01$), from a baseline of 25 percent. The estimate is stable in column (2) when provincial exposure is added, and paternal and province-level exposure effects on maternal wage employment are small and statistically insignificant. Because only one in four mothers held a wage job before the BTA, measuring wage levels directly would leave most of the sample without an observed market wage. The transition into wage employment from informal or agricultural work is itself a signal of improved labor market conditions, and the shift from the informal to the for-

mal sector is precisely the reallocation that McCaig and Pavcnik (2018) document at the aggregate level. Our empirical first stage therefore captures formalization rather than a directly measured wage increase. In the Vietnamese context, formalization tightens the maternal time constraint through the rigidity of formal work schedules, which limit the flexibility to combine childcare with market work.

On the intensive margin, columns (3)–(4) show that mothers in high-exposure industries work 0.15 log-points more monthly hours ($p < 0.01$), approximately a 15 percent increase from a baseline of about 110 hours per month.[18] Because the VHLSS provides only one pre-treatment wave (2002) for labor outcomes, formal pre-trend testing is infeasible for these specifications. We therefore present the labor market results as supportive reduced-form evidence for the maternal time-cost mechanism rather than stand-alone causal claims. The sex-selection event study (Figure 3), which uses richer pre-treatment timing from the 1999 Census, provides the primary parallel-trends validation. The hours estimates also require caution because the BTA shifts some mothers from informal or agricultural work into wage employment. For these women, recorded hours may reflect a new wage job rather than a change in total work effort. The scope for this compositional concern is limited because female labor force participation in Vietnam was already approximately 70 percent before the BTA (Section 3).

Columns (5)–(6) show the mirror pattern for fathers. Maternal industry exposure ($M_i \times \text{Post}_t$) is near zero for paternal wage employment, confirming that the BTA did not create spurious cross-parent effects. In contrast, paternal industry exposure ($F_i \times \text{Post}_t$) enters at 15 percentage points ($p < 0.01$), confirming that the BTA genuinely improved fathers' labor market conditions through their own industry. The asymmetry across columns is the central labor market result. Each parent's wage employment responds to their own industry exposure, but only the maternal channel transmits to sex selection.

5.3 Effects on Fertility

Predictions 1 and 2 generate opposing fertility effects. Maternal exposure reduces births through time costs, while paternal exposure raises births through income (Table 1, column 3). Table 5 reports estimates where the dependent variable is an indicator equal to one if the mother had any birth in the past 12 months.

The sample is a retrospective panel of woman-year observations with mother fixed effects, so identification comes from within-mother changes in birth propensity before and

[18] As noted in Section 4, the labor-supply specifications exclude birth parity because it is endogenous to labor market conditions. Adding parity as a control leaves the maternal coefficients essentially unchanged.

Table 5: Effects of BTA Exposures on Fertility (Probability of Giving Birth)

	AnyBirth					
	(1)	(2)	(3)	(4)	(5)	(6)
$M_i \times$ Post	-0.004***		-0.006***	-0.007***	-0.006***	-0.007***
	(0.001)		(0.001)	(0.001)	(0.001)	(0.001)
$F_i \times$ Post		0.003*	0.005***	0.005***	0.005***	0.005***
		(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
$P_p \times$ Post					-0.0001	-0.002
					(0.002)	(0.002)
R^2	0.29	0.29	0.29	0.29	0.29	0.29
Observations	2,011,963	2,011,963	2,011,963	2,011,963	2,011,963	2,011,963
Dep. Var. Mean	0.02	0.02	0.02	0.02	0.02	0.02
Controls				✓		✓

Notes: Dependent variable is an indicator for any birth in the past 12 months. Controls in columns (4) and (6) include mother's age, world-average tariff controls at parents' industries (birth-year matched), and pre-BTA provincial employment shares interacted with birth-year dummies. All specifications include mother and birth-year fixed effects. Province-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

after the BTA. Maternal industry exposure reduces the probability of giving birth. Column (1) yields a coefficient of -0.004 ($p < 0.01$) when maternal exposure enters alone, and the estimate ranges from -0.006 to -0.007 ($p < 0.01$) across the combined specifications in columns (3) through (6), confirming that the result does not depend on controlling for paternal or provincial income channels. Because the dependent variable mean is only 0.02, a reduction of 0.6 percentage points represents a 30 percent decline in the annual birth probability. The treatment combines an extensive-margin shift into wage employment and an intensive-margin increase in hours worked (Table 4), both of which create direct time conflicts with childbearing. The sign and magnitude are consistent with the model prediction that maternal industry exposure reduces fertility by raising the opportunity cost of the mother's time (Table 1, column 3, row 1).

Paternal industry exposure moves in the opposite direction. The father's coefficient is 0.003 ($p < 0.10$) when entered alone in column (2) and rises to 0.005 ($p < 0.01$) once maternal exposure is included in columns (3) through (5). The 0.005 estimate implies a 25 percent increase in annual birth probability relative to the mean. The positive sign is con-

sistent with the model prediction that paternal exposure raises fertility through an income effect without tightening the maternal time constraint (Table 1, column 3, row 2). The increase from column (2) to column (3) reflects the positive correlation between maternal and paternal exposures, so omitting maternal exposure attenuates the paternal estimate downward. The distinction between the two channels is thus not only a theoretical prediction but also an empirical regularity. Once each parent's exposure is held constant, the signs diverge sharply in the direction the model predicts.

Province-level exposure ($P_p \times \text{Post}_t$) is near zero and statistically insignificant in column (5). As with sex selection, the model predicts this channel should have the weakest effect, because non-labor income does not enter any effective price and is a small share of full household income. The provincial null also rules out the possibility that broad local economic movements are driving the fertility result.

The opposing signs of maternal and paternal exposure on fertility, combined with the null for provincial exposure, are difficult to reconcile with a model where any positive income shock affects childbearing in the same direction. The pattern instead supports the structural distinction at the core of our framework. In the model, maternal exposure tightens the relevant time constraint, reducing births and raising the incentive for sex selection, while paternal exposure relaxes the household budget, increasing births without affecting the sex composition of children. The opposing signs also speak to a natural objection, namely that higher income could ease childbearing by allowing households to purchase childcare. If purchased childcare could fully substitute for the mother's time, maternal exposure would resemble paternal exposure, raising fertility through income rather than reducing it through time costs. The fact that the two fertility effects point in opposite directions suggests that the maternal time constraint remains important and that childcare substitution is limited in this setting, consistent with Vietnam's scarce formal childcare outside major urban centers.

One qualification is that the 12-month birth indicator captures annual birth hazard rather than completed fertility, so part of the estimated reduction may reflect birth timing rather than a permanent decrease in desired family size. The heterogeneity estimates in Figure 4, which show larger effects among older mothers facing a narrowing fertility window, provide indirect evidence against a pure timing interpretation. The fertility estimate is also sensitive to the retrospective lookback window (Appendix Table C11), with the negative effect growing and stabilizing as the window expands. We therefore treat the fertility magnitude as less stable than the sex-selection result. The fertility results are consistent with the time-cost mechanism, showing that the same industry-level shock that

increases maternal labor supply also reduces births.[19]

An important qualification applies to the joint interpretation of the sex-selection and fertility results. The sex-selection estimate conditions on realized births. Because maternal exposure reduces annual birth probability by 30 percent, the composition of the birth sample may change across treatment groups, and if suppressed births differ systematically in sex composition, the observed male share among remaining births could shift mechanically. Among first births (Figure 4), where selective stopping cannot operate because the mother has not yet observed any child's sex, the maternal coefficient is 0.063 (s.e. 0.035, $p < 0.10$), the same sign as the full-sample estimate but smaller. This partial diagnostic does not point to stopping-rule-driven composition as the main explanation, but it does not address selection into having any birth after the BTA. We therefore interpret the sex-selection result as the effect on the male share among realized births, and treat the fertility decline as supportive mechanism evidence rather than a clean decomposition of the sex-selection effect.

5.4 Panel Evidence

A remaining concern with the cross-sectional estimates is that women with stronger son preference may have sorted into high-tariff-cut industries after the BTA, inflating the estimated maternal exposure coefficient. Table 6 addresses this concern by moving from the cross-sectional specification to a within-mother panel design. The progression is informative but not mechanical: column (2) changes both the sample and treatment assignment, column (3) replaces location and industry fixed effects with mother fixed effects, and column (4) adds additional channels and demographic controls. We interpret the panel results as reducing concern that time-invariant son preference or contemporaneous industry switching explains the cross-sectional estimate, while recognizing that they do not by themselves eliminate dynamic selection or other time-varying confounding.

Column (1) estimates the maternal exposure effect on the 2002–2006 cross-sectional sample. The maternal coefficient is 0.06 ($p < 0.05$), matching the full-sample estimate in Table 3. Column (2) then makes two substantive changes at once: it restricts the sample to panel-eligible mothers and assigns treatment based on the mother's 2002-wave industry rather than her contemporaneous industry. Among women working in 2002, approximately 23% changed their 2-digit ISIC industry by 2004 or 2006, so freezing treatment at the 2002 assignment reduces concern that contemporaneous post-BTA occupation

[19] Appendix Table C9 reports the same fertility specification with standard errors clustered at the mother level rather than the province level, to account for within-mother serial correlation in the retrospective panel. The results are unchanged.

Table 6: Panel Evidence on Within-Mother Identification

	MaleBirth			
	(1)	(2)	(3)	(4)
$M_i \times$ Post	0.06**	0.17**	0.34***	0.30***
	(0.03)	(0.07)	(0.08)	(0.11)
$F_i \times$ Post				0.06
				(0.16)
$P_p \times$ Post				0.03
				(0.06)
R^2	0.01	0.09	0.83	0.82
Observations	12,127	2,684	2,788	2,615
Mothers		1,723	1,723	1,723
Dep. var. mean	0.53	0.52	0.52	0.53
Individual FE			✓	✓
Industry FE	✓	✓		
Province FE	✓	✓		

Notes: Column (1) uses the full 2002–2006 cross-section with contemporaneous industry assignment. Columns (2)–(4) restrict to 1,723 panel mothers with 2002-wave industry assignment. Within-mother identification in columns (3)–(4) is driven by the subset of mothers with births both before and after the BTA. Column (3) adds mother fixed effects. Column (4) adds paternal and provincial channels. All columns include birth year FE and demographic controls. Appendix Table C1 compares the 2002-baseline subset of the rotating panel ($N = 879$) to the main cross-section. Province-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

changes mechanically generate the treatment effect. The maternal coefficient rises to 0.17 ($p < 0.05$). Table C1 shows that panel-eligible mothers are broadly similar to the broader cross-section on education, ethnicity, urban residence, and province-level tariff exposure, with the main difference being that they are about one year older on average. We therefore view column (2) as moving the design closer to pre-determined treatment assignment while staying close to the main sample on observables.

Column (3) replaces province and industry fixed effects with individual mother fixed effects, absorbing time-invariant maternal heterogeneity including baseline son preference and permanent industry sorting. The maternal coefficient rises to 0.34 ($p < 0.01$). Of the 1,723 panel mothers, 312 have births on both sides of the BTA and contribute 926

births to the within-mother identification. A leave-one-out exercise confirms that no single mother shifts the maternal coefficient by more than 15 percent, with estimates ranging from 0.29 to 0.38. The estimate is therefore not driven by a handful of influential observations.

The larger within-mother estimate relative to the cross-section likely reflects the selected panel sample of high-fertility women facing a narrowing fertility window, a group for whom the heterogeneity analysis (Figure 4) already shows larger treatment effects. We therefore use the panel for sign and identification direction rather than magnitude. The continuous-treatment translation at the median tariff cut, 1.5 to 2.7 percentage points, provides the more appropriate magnitude benchmark. The panel comparison strengthens the case against a narrow sorting story, but it does not by itself rule out dynamic selection or other time-varying confounding.

Column (4) adds paternal and provincial exposure channels and demographic controls. The maternal coefficient is essentially unchanged at 0.30 ($p < 0.01$). The stability confirms that M_i does not proxy for paternal or provincial income shocks. The paternal and provincial channels are individually small and statistically insignificant, consistent with the main results in Table 3.

Summary of the main results. Taken together, the cross-sectional and panel results trace a joint pattern across the three exposure channels and three outcomes that maps directly onto the model predictions in Table 1. Maternal industry exposure increases male birth probability, increases maternal labor supply, and reduces fertility, as Prediction 1 requires. Paternal industry exposure has at most weak effects on sex selection but raises wage employment and fertility, consistent with Prediction 2. Provincial exposure does not add a parallel sex-selection or fertility pattern once the parent-specific channels are included, consistent with Prediction 3. Section 7 tests whether this joint pattern can distinguish the maternal time-cost mechanism from the income, relative-returns, and bargaining alternatives identified in Prediction 4.

6. Robustness Checks

This section examines the sensitivity of our main results to alternative estimation choices. We consider alternative clustering schemes, region-by-year fixed effects, nonlinear specifications, and continuous treatment measures.

Alternative Clustering. Our baseline specifications cluster standard errors at the province level (61 clusters), the natural choice given that the BTA shock propagates through local labor markets and households. However, because treatment is defined by the tariff cut assigned to each parent's 2-digit ISIC industry, unobserved shocks may also correlate within industries. We address this concern with two alternative clustering schemes. Table C2 clusters by 59 2-digit ISIC industries alone, directly testing whether within-industry error correlation inflates our baseline inference. Table C3 applies two-way clustering by province and industry simultaneously, the most conservative approach as it accounts for both geographic and sectoral correlation in the error term.

Under ISIC-only clustering (Table C2), the maternal exposure coefficients on log hours (0.15), wage employment (0.15), and fertility (-0.007) all remain significant at the 1 percent level, with standard errors close to the baseline. The maternal effect on sex selection retains a similar magnitude (0.08) and is significant at the 1 percent level, though the standard error rises modestly from 0.02 to 0.03 under industry clustering. Under two-way clustering (Table C3), all four maternal exposure effects are significant at the 1 percent level, with standard errors close to the baseline for sex selection and labor supply, though modestly larger for fertility. Across both approaches, the paternal exposure coefficients remain small and insignificant for sex selection and maternal labor outcomes, though the paternal fertility effect (AnyBirth 0.005) retains significance at the 5 percent level. The provincial exposure coefficient for hours reaches the 5 percent threshold under ISIC-only clustering but is insignificant under two-way clustering. The overall pattern is consistent with the baseline conclusion that maternal industry exposure drives the main findings.

Region-Year Fixed Effects. A potential concern is that regional infrastructure improvements could independently affect access to sex-selective technologies (Chen et al., 2013) or childcare services, confounding our estimates of trade exposure effects. Vietnam's infrastructure development typically follows its eight administrative regions (Northeast, Northwest, Red River Delta, North Central Coast, South Central Coast, Central Highlands, Southeast, and Mekong River Delta), and shocks to healthcare access or childcare provision may vary across these regions over time. Since these regions correspond to official planning units, region-by-year fixed effects absorb region-level policy variation and differential investment in healthcare or childcare infrastructure. We incorporate region-by-year fixed effects into our baseline specifications to absorb any such time-varying regional confounders.

Table C4 shows that including region-by-year fixed effects leaves all four outcomes essentially unchanged. The maternal exposure coefficients retain their baseline magnitudes

and significance levels across sex selection (0.08, $p < 0.05$), hours (0.15, $p < 0.01$), wage employment (0.15, $p < 0.01$), and fertility (-0.007, $p < 0.01$). Paternal and provincial exposure measures remain small for sex selection and labor supply. The paternal fertility coefficient (0.005, $p < 0.01$) retains its significance, consistent with the main results. The results are unlikely to be driven by omitted regional shocks to infrastructure or healthcare access.

Nonlinear (Logit) Estimation. Given our binary outcome (sex selection), we check robustness to nonlinear specification using a fixed-effects logit model, which is more computationally feasible than probit given numerous fixed effects. Table C5 reports the logit estimates. The maternal exposure coefficient ($M_i \times$ Post) is 0.22 without controls and 0.33 with controls, both significant at the 5 percent level. The implied odds ratio from the controlled specification is $e^{0.33} \approx 1.39$, indicating a 39 percent increase in the odds of a male birth for mothers in high-exposure industries. The implied average marginal effect of approximately 0.08 is close to the linear probability model estimate, confirming that the linear probability approximation is accurate at the sample mean.

The paternal and provincial exposure coefficients remain small and statistically insignificant across both specifications, confirming that maternal industry exposure drives the sex-selection response. The agreement between logit and linear probability model results confirms that the main estimates are not an artifact of the linear functional form.

Continuous Treatment Exposures. Our main analysis uses binary indicators that classify workers as high-exposure or low-exposure based on whether their industry's tariff cut exceeds the sample median. This binary split discards variation in exposure intensity within each group. We now replicate the analysis using continuous tariff-cut measures ($\tau_m = \tau_{j(m)}$, $\tau_d = \tau_{j(d)}$, and τ_p from Section 3.1), which exploit the full distribution of tariff reductions across industries. Tables C6, C7, and C8 present estimates for sex selection, maternal labor supply, and fertility.

The sex-selection result is robust to the continuous specification. Table C6 shows that the maternal exposure coefficient on male birth ($\tau_m \times$ Post) is positive and significant in all specifications where it is included (columns 1 and 3–6). The coefficient ranges from 0.21 without controls to 0.38 with the full control set ($p < 0.01$), paralleling the pattern in the binary specifications where adding controls strengthens the estimate. Column (2) isolates the paternal channel and does not include τ_m. Paternal and provincial exposure coefficients are small and insignificant, as in the binary results.

The maternal labor supply results in Table C7 confirm the wage-employment effect

and remain directionally consistent on hours. The maternal exposure coefficient on wage employment is 0.51 without controls and 0.63 with the full control set ($p < 0.01$), closely matching the binary specifications. For hours, the maternal coefficient remains positive, ranging from 0.15 to 0.35 log points, but is imprecisely estimated under the continuous specification. We therefore treat the wage-employment result as the stronger continuous-treatment evidence on maternal labor supply, while viewing the hours estimates as supportive but less precise. Provincial exposure enters negatively for hours in one specification (-0.08, $p < 0.05$), but because this pattern does not appear in the binary design, we do not emphasize it.

For fertility, Table C8 confirms the negative maternal exposure effect. When maternal and paternal treatments are jointly included, the $\tau_m \times$ Post coefficient is -0.008 ($p < 0.05$), consistent with the binary estimate of -0.006 ($p < 0.01$). The attenuation of the unconditional maternal estimate (-0.003, column 1) toward zero reflects positive correlation between maternal and paternal exposure: paternal exposure raises fertility through an income effect, partially offsetting the maternal time-cost effect in a single-exposure regression. Controlling for paternal exposure sharpens the maternal estimate, producing coefficients between -0.008 and -0.012 that are statistically significant in columns (3)–(6). Paternal exposure ($\tau_d \times$ Post) consistently raises fertility, with coefficients ranging from 0.021 to 0.026 ($p < 0.01$) across all combined specifications, consistent with the binary result that $F_i \times$ Post increases birth probability. The provincial continuous exposure also enters positively and significantly for fertility in columns (5)–(6). Unlike the binary specification, the continuous specification exploits the full distribution of local exposure intensity, so this result is consistent with broader local-income effects of the trade shock operating through the provincial channel. At the same time, because this pattern does not appear in the binary specification, we treat it as suggestive rather than as a central result.

Across all three outcomes, the continuous specifications show that maternal labor supply responds to maternal industry conditions and are consistent with the view that the sex-selection effect operates through the maternal time-cost mechanism rather than a generalized household income channel. These continuous-treatment estimates should nonetheless be interpreted with caution, as standard TWFE regressions with a continuous treatment variable can assign negative implicit weights to some dose groups and require a substantially stronger parallel trends assumption (Section 3.1). The binary specifications serve as the identification workhorse for sign and significance, while the continuous-treatment specifications provide the appropriate magnitude benchmark.

7. Alternative Mechanisms

The pattern of results across outcomes and exposure channels allows us to distinguish the maternal opportunity-cost mechanism from three competing explanations. Maternal exposure increases sex selection, raises labor supply, and reduces fertility. Paternal exposure raises fertility but has no effect on sex selection. Provincial exposure is weak across all outcomes. Each alternative mechanism generates predictions that differ from this joint pattern on at least one outcome-channel combination, and Table 1 in Section 2 summarizes the relevant comparative statics.

Under the income-effects explanation (Almond et al., 2019), any income source should increase sex selection, including paternal wages and provincial income. Across all specifications, paternal and provincial exposure coefficients for sex selection are small and statistically insignificant (Tables 3 and C6), even though both channels produce meaningful effects on wage employment and fertility. The joint pattern is difficult to reconcile with a model in which income alone drives sex selection.

The relative-returns mechanism (Qian, 2008; Rosenzweig and Schultz, 1982) predicts that expanding opportunities in female-intensive industries should reduce sex selection by making daughters more valuable. The maternal exposure coefficient is positive across all specifications (Tables 3 and C6), contradicting this prediction.

The preceding test interprets the relative-returns channel at the level of individual wages. In the model of Section 2, returns to daughters' human capital equal the maternal wage. Through the cost channel, higher w_m raises sex selection, while the value channel works in the opposite direction (Section 2.2). The net positive effect documented in our estimates suggests the cost channel dominates. But the relative-returns mechanism's original argument operates at a different level. It is not about whether a particular mother earns more today, but whether daughters born now will face better economic prospects as adults in the local labor market. When female-intensive industries expand in a province, all families in that province have reason to expect that daughters will find more rewarding work when they grow up, regardless of the mother's own industry. Appendix A.2.4 generalizes the model to allow province-level returns to differ from individual wages. Under that extension, cost and value channels oppose each other: daughters become more expensive to educate at optimal quality (raising sex selection) but also more economically valuable (reducing son preference). The net effect is ambiguous.

To test this local-market reading directly, we construct a female-weighted provincial BTA exposure measure $\tilde{P}_p^w = \sum_j f_j \cdot s_{jp} \cdot \tau_j$, where f_j is the pre-BTA female employment share in industry j, s_{jp} is the pre-BTA employment share of industry j in province p, and

τ_j is the industry-level tariff cut. Because $f_j \cdot s_{jp}$ need not sum to one across industries, $\tilde{P}_p^w$ captures both the intensity of tariff cuts in female-intensive industries and the baseline importance of female employment in each province. This measure is the relevant object for testing the relative-returns mechanism, since provinces where female-intensive industries are both larger and more exposed to tariff cuts are precisely those where daughters' future labor market prospects improve most. We define $P_p^w = \mathbf{1}[\tilde{P}_p^w > \text{median}]$ as a binary indicator for above-median female-weighted provincial exposure. Provinces where $P_p^w = 1$ are precisely those where the BTA most expanded demand in female-intensive industries, generating the variation in daughters' local economic value that the relative-returns mechanism emphasizes. Under this account, $P_p^w \times \text{Post}_t$ should reduce sex selection. We estimate:

$$y_{ipt} = \beta_1(M_i \times \text{Post}_t) + \beta_2(F_i \times \text{Post}_t) + \beta_3(P_p \times \text{Post}_t) + \beta_4(P_p^w \times \text{Post}_t) + \text{FE} + X'_{ipt}\phi + \varepsilon_{ipt}, \tag{17}$$

where the relative-returns prediction is $\beta_4 < 0$ for sex selection. Table C10 reports the results across all four outcomes, entering all exposure channels simultaneously. The female-weighted provincial exposure coefficient is small and statistically insignificant for sex selection and fertility. The one outcome where it is large and significant is maternal wage employment, where $P_p^w \times \text{Post}_t$ raises the probability that mothers hold wage jobs by 5 percentage points (column 3, $p < 0.05$). This reflects higher returns to female labor in provinces most exposed to female-intensive industry expansion, which is precisely the local labor-market condition that the relative-returns mechanism predicts should raise the economic value of daughters and reduce sex selection. The mechanism's first stage is thus present in the data: female labor market returns rose where the mechanism predicts they should. Yet higher returns do not reduce son preference. Neither the local-labor-market reading nor the individual-wage reading of the relative-returns mechanism accounts for the sex-selection patterns we observe.

The bargaining explanation presents a subtler challenge. The key empirical failure is immediate. If bargaining drives the maternal sex-selection effect, then paternal wage gains should shift bargaining power toward the father and reduce sex selection. Paternal BTA exposure does produce meaningful economic effects. Fathers in high-exposure industries are 15 percentage points more likely to hold wage jobs (Table 4), and paternal exposure raises the annual birth probability by 0.5 percentage points (Tables 5 and C10). Yet the paternal exposure coefficient for sex selection is small and statistically insignificant across the binary (Table 3), continuous (Table C6), and panel (Table 6) specifications. Higher paternal wages raised income and fertility, as the baseline model predicts, but did

not shift son preference.

The formal bargaining extension in Appendix A.3 confirms this logic. The extension allows bargaining weights to depend on relative wages, $\mu = \mu(w_m/w_d)$ with $\mu'(\cdot) > 0$, and permits heterogeneous preferences across spouses, α_{Bg} and α_{ng} for $g \in \{m, d\}$. Under the parameter restrictions that match the positive maternal effect on sex selection ($\alpha_{Bm}/\alpha_{nm} > \alpha_{Bd}/\alpha_{nd}$) and the negative maternal effect on fertility ($\alpha_{nm} < \alpha_{nd}$), the model predicts $\partial b/\partial w_d < 0$. The data reject this prediction.

A simpler test reaches the same conclusion without imposing bargaining structure. If the positive maternal effect on sex selection reflected mothers exercising son preference as their bargaining power grew, then mothers who already hold a stronger bargaining position should select more sons even before the trade shock. We test this prediction using the 1999 Census, before the BTA took effect. Among births in 1997 and 1998, mothers with strictly more education than their husbands, roughly one quarter of the sample, are no more likely to have a son (Table C12). The absence of any pre-treatment association between relative parental education and sex selection does not support the premise that maternal bargaining power drives son preference.

The specific prediction that fails is the link from paternal wage gains to reduced son preference. The null sex-selection effect is not simply underpowered, since we observe significant paternal effects on labor supply and fertility. The baseline model accounts for this pattern without requiring parameter restrictions on spousal preferences.

8. Concluding Remarks

The key finding of this paper is a joint pattern across exposure channels and outcomes. Maternal exposure to the 2001 US-Vietnam Bilateral Trade Agreement increases sex selection, raises labor supply, and reduces fertility. Paternal and provincial exposure channels are silent on sex selection. At the median industry tariff cut, mothers are approximately 2 percentage points more likely to have a male child, work approximately 15 percent more hours, and have a 30 percent lower annual birth probability. Panel analysis with individual fixed effects confirms the sign and direction of the cross-sectional result, reducing concern that sorting drives the finding.

Tests of competing mechanisms reveal a consistent pattern. Each alternative mechanism has an operative first-stage effect on wage employment and labor supply, but none transmits to sex selection. Paternal BTA exposure raises wage employment and fertility, as the income channel predicts, yet leaves sex selection unchanged. Female-weighted provincial exposure raises maternal wage employment but does not reduce son prefer-

ence, and paternal exposure does not shift sex selection in the direction the bargaining model requires. The maternal time-cost mechanism is the channel most consistent with the full set of results.

The mechanism we identify is not unique to Vietnam. Albania, Montenegro, and Georgia show similar trajectories around their own trade integration episodes (Figure 1), and the same structural conditions are present in many other developing economies with son preference, patrilocal family structures, and rapid expansion of female employment, including parts of Indonesia and sub-Saharan Africa. The key moderating conditions are not trade itself but the conjunction of maternal time costs and son preference. Where mothers bear the primary childcare burden and families desire sons, trade-induced female employment opportunities can simultaneously empower women economically and intensify demographic imbalances.

One limitation of our analysis is that we observe children aged 0–1 who are alive and co-resident at the time of the survey, so we cannot fully distinguish prenatal sex selection from postnatal mechanisms such as sex-differential infant mortality, non-co-residence, or selective fostering. Vietnam's infant mortality rate was low by regional standards around 2002, which limits the scope for postnatal channels to generate effects of the magnitude we document. Fostering of infants aged 0–1 is also rare in Vietnamese household structure. Administrative birth records linking ultrasound visits to maternal employment would provide a definitive test, but the postnatal concern appears more likely to be a measurement limitation than the main explanation for the cross-channel pattern. The effects concentrate among older mothers and in provinces with higher pre-existing sex ratios, suggesting that the mechanism is most operative where son preference is strongest and mothers face a narrowing fertility horizon.

Addressing the link between female employment and sex selection requires policies that reduce the maternal time constraint directly. Income transfers or wage subsidies alone will not suffice, since the underlying mechanism runs through time costs rather than income. What matters is expanding access to institutional childcare and other supports that decouple female labor market participation from the opportunity cost of raising children.

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A. Model Appendix

This appendix provides the full derivations supporting the model in Section 2. Appendix A.1 derives the first-order conditions and comparative statics for the baseline model via Slutsky decomposition. Appendix A.2 extends the model to include human capital investment in a quantity-quality framework. Appendix A.3 develops an endogenous bargaining extension.

A.1 Baseline Model Derivations

A.1.1 Lagrangian and First-Order Conditions

The household maximizes $U(c, \ell, n, B)$ with $B = bn$, choosing (c, ℓ, n, b) subject to the budget constraint $c + w_m \ell + w_m \gamma\, n + p_b\, b\, n = W$. The Lagrangian is

$$\mathcal{L} = U(c, \ell, n, bn) + \lambda \left[W - c - w_m \ell - w_m \gamma\, n - p_b\, b\, n \right]. \tag{18}$$

The first-order conditions with respect to c, ℓ, n, and b are

$$U_c = \lambda, \tag{19}$$

$$U_\ell = \lambda\, w_m, \tag{20}$$

$$U_n + U_B\, b = \lambda \left(w_m \gamma + p_b\, b \right), \tag{21}$$

$$U_B\, n = \lambda\, p_b\, n. \tag{22}$$

The fertility condition (Equation 21) includes $U_B b$ on the left (the marginal benefit of one more child through the boy channel, since each child yields b expected boys) and $p_b b$ on the right (the sex-selection cost per child). The sex-selection condition (Equation 22) simplifies to $U_B = \lambda p_b$, equating the marginal utility of boys to their shadow price.

Substituting $U_B = \lambda p_b$ from Equation (22) into Equation (21):

$$U_n + \lambda p_b b = \lambda w_m \gamma + \lambda p_b b \quad \Longrightarrow \quad U_n = \lambda w_m \gamma. \tag{23}$$

The $p_b b$ terms cancel. At the optimum, the effective shadow price of fertility is $w_m \gamma$, the maternal time cost alone. The household simultaneously optimizes b, so the sex-selection cost of an additional child ($p_b b$) is exactly offset by the marginal boy value ($U_B b$). Dividing through by $U_c = \lambda$, the MRS conditions reported in the main text (Equation 4) follow.

A.1.2 Slutsky Decomposition

The effective price structure $(1, w_m, w_m\gamma, p_b)$ for goods (c, ℓ, n, B) permits standard Slutsky decomposition.

Maternal wage effect on sex selection. The maternal wage w_m enters two prices, $p_\ell = w_m$ and $p_n = w_m\gamma$. The compensated effect on the sex ratio $b = B/n$ operates through both channels:

$$\frac{\partial b^S}{\partial w_m} = \underbrace{1 \cdot \frac{\partial b^S}{\partial p_\ell}}_{\text{leisure-price channel}} + \underbrace{\gamma \cdot \frac{\partial b^S}{\partial p_n}}_{\text{fertility-price channel}} > 0, \tag{24}$$

where positivity holds when n and B are Hicksian substitutes (the household substitutes from quantity toward boys when children become more expensive). The total (Marshallian) effect adds the income effect. Denoting the mother's labor supply $h_m = 1 - \ell - \gamma n$,

$$\frac{\partial b}{\partial w_m} = \underbrace{\frac{\partial b^S}{\partial w_m}}_{>0\text{ (substitution)}} + \underbrace{h_m \left(\frac{1}{n}\frac{\partial B}{\partial W} - \frac{b}{n}\frac{\partial n}{\partial W} \right)}_{\text{income effect on } b}. \tag{25}$$

The income effect on b depends on the relative income elasticities of B and n. When $\eta_B^W > \eta_n^W$ (boys are income-elastic relative to children), the income effect is positive, reinforcing the substitution effect. When $\eta_B^W = \eta_n^W$, the income effect vanishes. In either case, $\partial b/\partial w_m > 0$.

Maternal wage effect on fertility. Both price channels reduce fertility:

$$\frac{\partial n}{\partial w_m} = \underbrace{\frac{\partial n^S}{\partial w_m}}_{<0\text{ (substitution)}} + \underbrace{h_m \frac{\partial n}{\partial W}}_{\text{income effect}}. \tag{26}$$

If children are normal goods ($\partial n/\partial W > 0$), the income effect partially offsets the substitution effect. When the substitution effect dominates, $\partial n/\partial w_m < 0$.

Maternal labor supply. The time constraint $h_m = 1 - \ell - \gamma n$ pins down labor supply as a residual:

$$\frac{\partial h_m}{\partial w_m} = -\frac{\partial \ell}{\partial w_m} - \gamma \frac{\partial n}{\partial w_m}. \tag{27}$$

Both terms are positive when leisure and fertility fall, so $\partial h_m/\partial w_m > 0$.

Paternal wage effect. The paternal wage w_d enters no price. Its effect operates entirely through full income W:

$$\frac{\partial b}{\partial w_d} = \frac{1}{n}\frac{\partial B}{\partial W} - \frac{b}{n}\frac{\partial n}{\partial W}. \tag{28}$$

When $\eta_B^W > \eta_n^W$, the income effect on b is positive, giving $\partial b/\partial w_d > 0$. Because the father works full-time and typically earns more than the mother, the paternal income effect on b may exceed the maternal income effect. The critical asymmetry is that the maternal wage adds a large substitution effect (Equation 24) on top of the income effect, while the paternal wage has income only. The paternal wage also raises fertility ($\partial n/\partial w_d > 0$) and reduces maternal hours ($\partial h_m/\partial w_d < 0$). Non-labor income I works identically.

A.2 Human Capital Extension

This subsection derives the first-order conditions and comparative statics for the quantity-quality extension, then generalizes to allow province-level returns to differ from individual wages.[20]

A.2.1 The Quantity-Quality Lagrangian

The household chooses $(c, \ell, n, b, e_b, e_g)$ to maximize $U(c, \ell, n, B) + V(\bar{E})$ subject to the budget constraint in Equation (8), where $B = bn$, $\bar{E} = b\, r_b\, h(e_b) + (1-b)\, r_g\, h(e_g)$, and V is increasing and concave. The Lagrangian is

$$\mathcal{L} = U(c, \ell, n, B) + V(\bar{E}) - \lambda\big[c + w_m \ell + n(w_m \gamma + p_e(b\, e_b + (1-b)\, e_g)) + p_b\, b\, n - W\big]. \tag{29}$$

The first-order conditions for education investment are

$$\frac{\partial \mathcal{L}}{\partial e_b} = V'(\bar{E}) \cdot b\, r_b\, h'(e_b) - \lambda\, n\, b\, p_e = 0, \tag{30}$$

$$\frac{\partial \mathcal{L}}{\partial e_g} = V'(\bar{E}) \cdot (1-b)\, r_g\, h'(e_g) - \lambda\, n\, (1-b)\, p_e = 0. \tag{31}$$

Dividing by b and $(1-b)$ respectively and equating yields the education condition stated in the main text:

$$r_b\, h'(e_b^*) = r_g\, h'(e_g^*). \tag{32}$$

Because $h'' < 0$, the gender with higher returns receives more investment. When $w_d > w_m$, $e_b^* > e_g^*$.

[20] Diminishing returns ($h'' < 0$) are needed for the return to affect how much parents invest. Under linear h, optimal investment does not depend on r_g.

The first-order condition for b is

$$U_B \cdot n + V'(\bar{E}) \cdot \left[r_b\, h(e_b) - r_g\, h(e_g)\right] = \lambda\, n\, \left[p_b + p_e(e_b - e_g)\right]. \tag{33}$$

The left-hand side is the marginal benefit of raising b. The first term ($U_B \cdot n$) is the direct preference for sons. The second term captures how shifting the sex composition toward boys changes average human capital. When sons are more valuable than daughters ($r_b\, h(e_b) > r_g\, h(e_g)$), this term is positive, adding to the incentive for sex selection. The right-hand side is the marginal cost: the sex-selection technology cost p_b plus the education premium $p_e(e_b - e_g)$ for each son over each daughter, summed over n children.

A.2.2 The Quantity-Quality Interaction

From Equation (30), dividing by b:

$$V'(\bar{E}) \cdot r_b\, h'(e_b) = \lambda\, n\, p_e. \tag{34}$$

From Equation (31), dividing by $(1 - b)$:

$$V'(\bar{E}) \cdot r_g\, h'(e_g) = \lambda\, n\, p_e. \tag{35}$$

These expressions show that sex composition does not enter the education FOCs directly once we scale by the relevant child share. It enters indirectly through the shadow value λ. When $r_b > r_g$, optimal investment satisfies $e_b^* > e_g^*$. Holding (n, e_b^*, e_g^*) fixed, the education bill is $C_E = np_e[be_b^* + (1-b)e_g^*]$, so

$$\frac{\partial C_E}{\partial b} = np_e(e_b^* - e_g^*) > 0.$$

Replacing a daughter with a son therefore raises total education spending. The budget tightens, λ rises, and concavity of h implies that both e_b^* and e_g^* must fall at the new optimum. A higher male share b reduces schooling per child of both genders, not just per girl. Because n also appears on the right-hand side, the optimal education levels depend on fertility as well. When n falls, the shadow cost of quality decreases, and parents raise schooling investment. Symmetrically, the first-order condition for n includes education costs $p_e(b\, e_b + (1-b)\, e_g)$, so higher investment per child raises the cost of an additional child. This two-way dependence is the quantity-quality tradeoff of Becker and Lewis (1973).

A.2.3 Effective Prices and Comparative Statics

Rewriting the budget as in Equation (10), the effective prices are

$$p_c = 1, \qquad p_\ell = w_m, \qquad p_n = w_m\gamma + p_e\, e_g, \qquad p_B = p_b + p_e(e_b - e_g). \tag{36}$$

Relative to the baseline (where $p_n = w_m\gamma$ and $p_B = p_b$), the maternal wage now enters p_n through two channels and the paternal wage enters p_B through the son education cost differential. The comparative statics follow the Slutsky decomposition of Appendix A.1, applied to these modified prices. Because e_b and e_g are endogenous, they respond to changes in w_m, w_d, and n through the quantity-quality interaction (Equation 35).

How w_m affects the effective prices. With $r_g = w_m$, applying the implicit function theorem to the education condition (Equation 32) yields

$$\frac{de_g^*}{dw_m} > 0. \tag{37}$$

The education condition $r_b\, h'(e_b^*) = r_g\, h'(e_g^*)$ pins down the ratio of marginal products: $h'(e_b^*)/h'(e_g^*) = r_g/r_b$. When $r_g = w_m$ rises, this ratio increases, requiring a rise in e_g^* (to lower $h'(e_g^*)$) and potentially a fall in e_b^* (to raise $h'(e_b^*)$). In addition, the shadow cost of quality $\lambda n p_e$ shifts as the household reoptimizes n and λ, moving both investment levels. On net, the education gap $e_b^* - e_g^*$ narrows, because the direct effect on e_g^* through r_g is the leading force.[21] The effect on each effective price follows:

$$\frac{dp_n}{dw_m} = \gamma + p_e \frac{de_g^*}{dw_m} > \gamma, \tag{38}$$

$$\frac{dp_B}{dw_m} = p_e \left(\frac{de_b^*}{dw_m} - \frac{de_g^*}{dw_m} \right). \tag{39}$$

The fertility price p_n rises by a larger magnitude than in the baseline (where $dp_n/dw_m = \gamma$), because daughters' education costs increase alongside time costs. Both the direct and indirect channels reinforce this increase. The sex-selection price p_B likely declines, because the education gap $e_b^* - e_g^*$ narrows as daughters' education becomes more valuable.

[21] Holding the shadow cost of quality $\lambda n p_e$ fixed, the education condition alone gives $\partial e_g^*/\partial w_m > 0$ and $\partial e_b^*/\partial w_m = 0$, since only $r_g = w_m$ appears in the condition. In general equilibrium, λ and n also adjust, creating indirect effects on both e_b^* and e_g^* through the Becker and Lewis (1973) endogenous quality-price channel. The total derivative de_b^*/dw_m is therefore nonzero but small relative to de_g^*/dw_m, because the indirect channel operates on both investments symmetrically while the direct channel operates only on e_g^*.

The sign of dp_B/dw_m depends on the gap narrowing, which requires that the direct effect on e_g^* dominates the indirect effects on both investments. This condition holds when the education condition is the primary determinant of investment levels relative to the shadow-cost feedback.

Maternal wage effect on sex selection. In the baseline model, w_m enters only through prices in the budget constraint, and the Slutsky decomposition fully captures the substitution effect. With the human capital extension, w_m also enters the utility function through $r_g = w_m$ in $V(\bar{E})$. The maternal wage therefore affects b through two distinct routes. The first route operates through the budget constraint: w_m changes the effective prices, and the household reoptimizes along the budget. The Slutsky decomposition captures this price-side effect, which parallels Equation (24) with the modified prices:

$$\left.\frac{\partial b^S}{\partial w_m}\right|_{\text{HC}} = \underbrace{1 \cdot \frac{\partial b^S}{\partial p_\ell}}_{\text{leisure price}} + \underbrace{\frac{dp_n}{dw_m} \cdot \frac{\partial b^S}{\partial p_n}}_{\text{fertility price (amplified)}} + \underbrace{\frac{dp_B}{dw_m} \cdot \frac{\partial b^S}{\partial p_B}}_{\text{sex-selection price (new)}} . \tag{40}$$

In the baseline model, only the first two terms appear. The human capital extension adds the education component to the fertility-price channel and introduces the third term. We call the entire price-side mechanism the *cost channel*. Rising w_m makes children more expensive to raise and sons relatively cheaper, pushing b up. The first two terms of the cost channel are positive under the baseline condition that fertility and boys are Hicksian substitutes ($\partial b^S/\partial p_n > 0$). The third term is positive when the compensated own-price effect on boys is negative ($\partial b^S/\partial p_B < 0$) and the education gap narrows ($dp_B/dw_m < 0$). The own-price condition means that when sons become more expensive the household selects fewer of them. The gap-narrowing condition requires that the direct effect on daughters' education dominates the indirect shadow-cost feedback, as discussed above. When both conditions hold, all three cost-channel forces reinforce.

The second route operates through the utility function, not the budget. Because $r_g = w_m$, the maternal wage also enters $V(\bar{E})$ directly. This route is not captured by the Slutsky decomposition above, which only handles price effects. From the b-FOC (Equation 33), the marginal benefit of sex selection includes $V'(\bar{E}) \cdot [r_b\, h(e_b) - r_g\, h(e_g)]$. When w_m rises, r_g increases, so daughters' human capital $r_g\, h(e_g^*)$ rises and the gap $r_b\, h(e_b) - r_g\, h(e_g)$ narrows. The marginal benefit of selecting for sons falls. We call this the *value channel*. When daughters' future earnings prospects improve, each daughter becomes more valuable, and the household has less reason to skew births toward sons.

The total effect of w_m on b combines both routes:

$$\frac{db}{dw_m} = \underbrace{\left.\frac{\partial b^S}{\partial w_m}\right|_{\text{HC}}}_{\substack{\text{cost channel} \\ (+,\text{ through prices})}} + \underbrace{\left.\frac{\partial b}{\partial r_g}\right|_{\text{prices fixed}}}_{\substack{\text{value channel} \\ (-,\text{ through } V)}} + \text{income effect.} \tag{41}$$

The cost channel is positive (Equation 40), with three reinforcing forces: the leisure price effect, the amplified fertility price effect, and the new sex-selection price effect. The income effect is weakly positive under the same conditions as the baseline (Section A.1). The value channel is negative. The cost channel pushes b up through multiple reinforcing price effects, and the income effect reinforces the same direction, while only the value channel pushes b down. The net effect is likely positive.

Maternal wage effect on fertility (amplified). In the baseline, $dn/dw_m < 0$ when the substitution effect dominates the income effect. The human capital extension strengthens the substitution effect because the fertility price rises from $p_n = w_m\gamma$ to $p_n = w_m\gamma + p_e e_g$, and $dp_n/dw_m > \gamma$ (Equation 38).

Paternal wage effect on sex selection (ambiguous). As with the maternal wage, w_d affects b through both prices and utility. On the price side, w_d enters p_B directly through sons' returns and p_n indirectly through the shadow-cost feedback:

$$\frac{dp_B}{dw_d} = p_e\left(\frac{de_b^*}{dw_d} - \frac{de_g^*}{dw_d}\right) > 0, \qquad \frac{dp_n}{dw_d} = p_e\frac{de_g^*}{dw_d} \geq 0. \tag{42}$$

Higher w_d raises sons' returns ($r_b = w_d$), increasing e_b^* and widening the cost gap $e_b - e_g$, so $dp_B/dw_d > 0$. The fertility price p_n also rises through the indirect channel: higher w_d raises full income, lowers the shadow cost of quality, and increases both e_b^* and e_g^*. This pattern is the Becker and Lewis (1973) endogenous quality-price feedback.[22] Because w_d enters p_B directly, the cost channel has a primary negative force through p_B and a secondary negative force through p_n:

$$\left.\frac{\partial b^S}{\partial w_d}\right|_{\text{HC}} = \underbrace{\frac{dp_B}{dw_d}\cdot\frac{\partial b^S}{\partial p_B}}_{<0\text{ (sons costlier)}} + \underbrace{\frac{dp_n}{dw_d}\cdot\frac{\partial b^S}{\partial p_n}}_{>0\text{ (quality-price feedback)}}. \tag{43}$$

[22] Holding the shadow cost of quality fixed, $de_g^*/dw_d = 0$ because $r_g = w_m$ does not depend on w_d, and $dp_n/dw_d = 0$. The positive total derivative operates entirely through the indirect channel.

The first term dominates: sons become more expensive to educate, discouraging sex selection. The second term, operating through the indirect quality-price channel, is positive but second-order. On the utility side, w_d enters $V(\bar{E})$ through $r_b = w_d$. When w_d rises, sons' human capital $r_b h(e_b^*)$ increases, widening the gap $r_b h(e_b) - r_g h(e_g)$ in the b-FOC (Equation 33). The marginal benefit of selecting for sons rises. This value channel is positive, the mirror image of the maternal case. The income effect is also weakly positive (Equation 28). The total effect is therefore:

$$\frac{db}{dw_d} = \underbrace{\left.\frac{\partial b^S}{\partial w_d}\right|_{\text{HC}}}_{\substack{\text{cost channel} \\ (-)}} + \underbrace{\left.\frac{\partial b}{\partial r_b}\right|_{\text{prices fixed}}}_{\substack{\text{value channel} \\ (+)}} + \text{income effect } (+). \tag{44}$$

Under the same reasoning as the maternal case, where the cost channel dominates the value channel, the cost channel dominates here as well. The human capital extension therefore adds a net negative force on sex selection from the paternal side. In the baseline, the paternal wage had only a weakly positive income effect on b. The human capital extension introduces an offsetting negative substitution effect through p_B, pulling the total paternal effect toward zero.

Paternal wage effect on fertility. The fertility prediction $\partial n / \partial w_d > 0$ is likely positive. The direct income effect raises fertility, as in the baseline. The human capital extension introduces a partially offsetting force: higher w_d raises p_n through the indirect quality-price channel (Equation 42), creating a substitution penalty on quantity. Because the direct income effect is the leading force and the indirect quality-price feedback is second-order, the net effect on fertility is likely positive.

A.2.4 When Returns Differ from Individual Wages

The main text sets $r_g = w_m$ and $r_b = w_d$, identifying children's returns with same-sex parental wages. This subsection relaxes that restriction. When female-intensive industries expand in a province, the local return to daughters' human capital may rise for all families, regardless of individual mothers' industries. Let r_g now denote this province-level return, which may differ from any individual mother's wage w_m.

The education FOCs carry through with r_g and r_b as separate parameters. The education condition becomes $r_b\, h'(e_b^*) = r_g\, h'(e_g^*)$, and the quantity-quality interaction (Equation 35) still determines optimal investment, so e_g^* depends on both r_g and n. The modified prices become $p_n = w_m \gamma + p_e\, e_g(r_g, n)$ and $p_B = p_b + p_e(e_b(r_b, n) - e_g(r_g, n))$. The

key difference from the individual-level case is that w_m still determines the time cost of childcare ($w_m\gamma$ in p_n), but r_g now independently determines the education investment in daughters.

When r_g rises independently of w_m, two opposing forces emerge. The cost channel raises sex selection: higher r_g increases optimal investment e_g^*, raising the fertility price p_n and lowering the sex-selection price p_B. The value channel lowers sex selection: daughters' economic value rises in $\bar{E}$, reducing the household's willingness to select for sons (through the b-FOC, Equation 33). Unlike the maternal wage case, there is no time-cost force ($w_m\gamma$) giving the cost channel a built-in advantage, because r_g moves independently of w_m. The net effect is therefore more genuinely ambiguous:

$$\frac{db}{dr_g} = \underbrace{\left.\frac{\partial b}{\partial r_g}\right|_{\text{cost}}}_{>0} + \underbrace{\left.\frac{\partial b}{\partial r_g}\right|_{\text{value}}}_{<0}. \tag{45}$$

This decomposition motivates the empirical test using female-weighted provincial exposure P_p^w in Section 7. The measure P_p^w captures province-level variation in returns to female labor that operates independently of individual maternal wages. If the cost and value channels approximately cancel, P_p^w would affect daughters' education and maternal labor supply (through e_g^* and the labor market) without affecting sex selection. The maternal wage effect (where $r_g = w_m$) can still be positive because the baseline time-cost channel adds a dominant force that the province-level measure lacks.

A.2.5 Binding Fertility Constraints

The baseline model treats fertility as a free choice variable. When policy constrains fertility, as under China's one-child policy, the quantity margin shuts off and the model's predictions change. Suppose n is fixed at $\bar{n}$. The household chooses (c, ℓ, b, e_b, e_g) to maximize $U(c, \ell, \bar{n}, B) + V(\bar{E})$ subject to

$$c + w_m\ell + \bar{n}\left[w_m\gamma + p_e(b\,e_b + (1-b)\,e_g)\right] + p_b\,b\,\bar{n} = W. \tag{46}$$

The education FOCs are unchanged. The education ratio (Equation 32) and the quantity-quality interaction (Equation 35) both carry through with $\bar{n}$ replacing n. The critical difference is that $\bar{n}$ is a parameter, not a choice variable. From Equation (35), the shadow cost of quality is $\lambda\bar{n}p_e$, which is lower when $\bar{n}$ is small. Investment per child is therefore a decreasing function of $\bar{n}$. When the constraint binds at a low level (such as one child), parents invest heavily in each child, and the education cost gap $p_e(e_b^* - e_g^*)$ is large.

The constrained model generates three predictions that differ from the unconstrained case.

First, the composition of the maternal wage effect on sex selection changes. In the unconstrained model, the fertility-price channel ($\gamma \cdot \partial b^S / \partial p_n$) is the primary positive force driving b up when w_m rises. When n is fixed at $\bar{n}$, the time cost $w_m \gamma \bar{n}$ becomes a lump sum in the budget. Because changing b does not alter the total childcare burden, the fertility-price channel is shut off. The remaining positive forces are the leisure-price channel, the sex-selection price channel (from the education gap), and the income effect. Whether the constrained maternal wage effect is larger or smaller than the unconstrained effect depends on the relative magnitude of these remaining forces versus the fertility-price channel that is lost. The net effect on b remains positive but is not necessarily stronger.

Second, the relative-returns channel is also amplified. When province-level returns r_g rise independently of w_m (as in the preceding subsection), the cost and value channels still oppose each other (Equation 45). Under constrained fertility, investment per child is high, so the return to a daughter $r_g\, h(e_g^*)$ is large. A marginal increase in r_g therefore produces a large increase in daughters' economic value. This amplifies the value channel relative to the cost channel, tilting the balance toward a negative effect of r_g on sex selection.

Third, the income effect on sex selection may become positive. In the unconstrained model, non-labor income I has a negligible effect on b because it does not enter any effective price (Table 1). Under constrained fertility, the income gain that would have raised n cannot do so. If sons are income-elastic at the margin ($\eta_B^W > \eta_n^W$), the blocked fertility response redirects into higher b.

Section 2 discusses how these predictions connect to findings from the China sex-selection literature.

A.3 Endogenous Bargaining Extension

Rising female wages may shift bargaining power within the household, affecting fertility decisions through a channel distinct from the time-cost mechanism. If mothers and fathers have different preferences over fertility and child gender, then changes in bargaining weights could generate maternal wage effects on sex selection even without asymmetric childcare costs.

Each parent has potentially different preferences over fertility and sons. Because the father works full-time ($h_d = 1$), his utility does not include a leisure term:

$$U_m = \alpha_c \log(c_m) + \alpha_\ell \log(\ell) + \alpha_{nm} \log(n) + \alpha_{Bm} \log(B), \tag{47}$$

$$U_d = \alpha_c \log(c_d) + \alpha_{nd} \log(n) + \alpha_{Bd} \log(B). \tag{48}$$

We normalize $\alpha_\ell + \alpha_{nm} + \alpha_{Bm} = \alpha_{nd} + \alpha_{Bd}$, so that the sum of non-consumption weights in each parent's utility is equal and the Cobb-Douglas expenditure-share denominator in the weighted objective is independent of μ.[23] The household maximizes a weighted sum of utilities where bargaining weights depend on relative wages:

$$\max_{\{c_m, c_d, n, b\}} \mu(w_m/w_d)\, U_m + (1 - \mu(w_m/w_d))\, U_d \tag{49}$$

subject to the budget constraint:

$$c_m + c_d + p_b\, b\, n \leq I + w_d + w_m\,(1 - \ell - \gamma\, n)\,, \tag{50}$$

where $\mu'(\cdot) > 0$ so higher relative maternal wages increase the mother's bargaining power. Time constraints remain as in the baseline model.

With log utility, the model yields two key reduced-form equations. The sex ratio at birth is:

$$\log b = \log\left(\frac{\mu\alpha_{Bm} + (1-\mu)\alpha_{Bd}}{\mu\alpha_{nm} + (1-\mu)\alpha_{nd}}\right) + \log\gamma - \log p_b + \log w_m. \tag{51}$$

Total fertility is:

$$\log n = \log\left(\frac{\mu\alpha_{nm} + (1-\mu)\alpha_{nd}}{\gamma}\right) + \log W - \log w_m. \tag{52}$$

Mother's wages increase sex selection ($\partial b/\partial w_m > 0$) if mothers have stronger relative son preference ($\alpha_{Bm}/\alpha_{nm} > \alpha_{Bd}/\alpha_{nd}$). Under this condition, maternal wages also reduce fertility ($\partial n/\partial w_m < 0$) because the $-\log w_m$ term captures the direct opportunity-cost effect. If mothers prefer fewer children than fathers ($\alpha_{nm} < \alpha_{nd}$), then father's wages increase fertility ($\partial n/\partial w_d > 0$) by shifting bargaining weight toward the higher-fertility parent. However, under these same parameter restrictions, the model also predicts that father's wages would reduce sex selection ($\partial b/\partial w_d < 0$), because higher paternal wages shift bargaining power toward fathers, who have weaker relative son preference by assumption. The bargaining model therefore predicts a negative paternal effect on b, whereas the time-cost model predicts a near-zero paternal effect. Section 7 tests this distinguishing prediction.

Adding human capital investment to the bargaining model does not change these conclusions. The quality valuation $V(\bar{E})$ is common to both parents, so it enters the weighted

[23] The mother's non-consumption weights sum to $\alpha_\ell + \alpha_{nm} + \alpha_{Bm}$ and the father's to $\alpha_{nd} + \alpha_{Bd}$. The weighted denominator is $\mu(\alpha_c + \alpha_\ell + \alpha_{nm} + \alpha_{Bm}) + (1-\mu)(\alpha_c + \alpha_{nd} + \alpha_{Bd})$, which is constant in μ under this normalization.

objective without interacting with the bargaining weight μ. The education FOCs (Equations 30–31) and the quantity-quality interaction (Equations 34–35) carry through unchanged. The only modification is that effective prices become $p_n = w_m\gamma + p_e e_g$ and $p_B = p_b + p_e(e_b - e_g)$, as in the unitary model with human capital. The human capital channel therefore reinforces the maternal effect and adds a negative force to the paternal effect, the same qualitative pattern as in Section 2.2. The bargaining and human capital channels are additive.

B. Data Appendix: Family Interrelationship Algorithm

Variables categorizing family interrelationships within the household are crucial in this paper. In particular, we need "pointer" variables identifying each person's mother, father, and spouse. VHLSS has a variable that classifies each individual's relationship to the household head. However, this variable is ambiguous and erroneous for the following reasons. First, extended family is common in Vietnam: roughly 34.91% of the VHLSS sample lives in the same household with relatives other than their nuclear family members. Because VHLSS does not identify subfamilies within the extended household, the "relate-to-head" variable will miss the interrelationship of subfamilies other than the head's. Second, with the exception of VHLSS 2002, all other waves do not identify "children-in-law." An immediate consequence is missing spousal links and therefore parental links in many cases. Thus, we write an algorithm to improve the precision of parent-child and spousal pairings in VHLSS.

We adapt a similar algorithm that has been used by the Minnesota Population Center (Sobek and Kennedy, 2009) to generate "pointer" variables for IPUMS–International Census Data. We utilize the four variables: relationship to household head, age, and marital status, in combination with the relative position of household members in the roster listing, to infer relationships. We first establish spousal links (generate variable SPLOC); then we find the mother and father for each individual (generate variables MOMLOC and POPLOC). Regarding parental linkage, we look for the mother before looking for the father. Once we find the mother for a child, we assign her husband as the father of this child. Only when we cannot find a child's mother do we locate its father (and assign his wife as the mother).

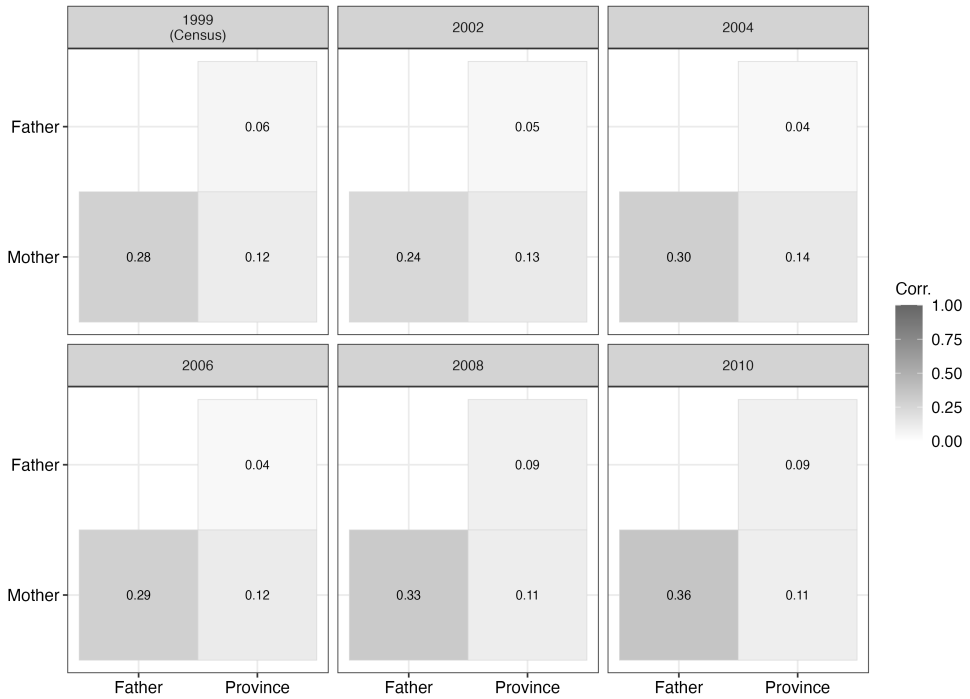
Tables B1 and B2 document the rules we apply to these matching tasks. Our algorithm applies the rules sequentially. If the first rule finds a match for a given individual, the second rule no longer applies to that person, and so on. Whenever there are ambiguous multiple potential spouses or multiple potential parents, we drop the entire household

from the sample.

Fortunately, VHLSS 2014 and 2016 provide two variables to locate the biological fathers and mothers of children under 16 years old. We use this information to test our algorithm and report the results in Table B3. The algorithm finds both parents correctly for 88.95% of the children, the correct mother (regardless of father match) for 93.23% (88.95 + 4.28), and the correct father (regardless of mother match) for 90.99%. The correction rate for both biological parents is lower than that for each biological parent because our algorithm also counts stepmothers and stepfathers. With these results, we are very confident that our algorithm does a good job at identifying family interrelationships.

C. Additional Tables and Figures

Figure C1: Correlations between Treatment Variables by Survey Wave



Notes: Pairwise correlations between tariff exposure measures by survey wave. Mother (τ_m) = maternal industry exposure, Father (τ_d) = paternal, Province (τ_p) = shift-share provincial. Data from the 1999 Census (baseline) and VHLSS (2002–2010).

Table B1: Rules for SPLOC Construction

Rule	Individual's relationship to head	Partner's relationship to head	Age difference	Both Married	Require adjacency	Only applicable to 2002	Notes
Strong couple pairing, couple adjacency preferred							
	Head	Spouse	No	No	No		1
	Parent	Parent	No	Yes	Yes		1
	Grandparent	Grandparent	No	Yes	Yes		1
	Child	Child-in-law	No	Yes	Yes	Yes	1
Weak couple pairing, couple adjacent							
	Grandchild	Grandchild	Yes	Yes	Yes		1, 2
	Other relationship	Other relationship	Yes	Yes	Yes		1, 2
	Sibling	Sibling	Yes	Yes	Yes	Yes	1, 2
	Grandchild	Other relationship	Yes	Yes	Yes		1, 2
	Sibling	Other relationship	Yes	Yes	Yes	Yes	1, 2
	Child	Other relationship	Yes	Yes	Yes		1, 2
Weak couple pairing, special type child-child							
	Child	Child	Yes	Yes	Yes		1, 2, 6
Weak couple pairing, couple not adjacent							
	Child	Child-in-law	No	Yes	Closest proximity	Yes	5
	Child	Other relationship	Yes	Yes	No		2, 3
	Child	Child	Yes	Yes	No		1, 2, 4, 6

Notes:

1. Drop the entire household if there is any person that could be assigned to 2 couples by the adjacency rule.
2. A woman can be no more than 20 years older or 35 years younger than a potential male partner.
3. For non-adjacent couple pairing, among the potential spouses who satisfy the age and marital requirements, select the person who is closest in age and impose that the husband is older than the wife.
4. For child-child non-adjacent couple pairing, drop all households where this rule yields multiple potential spouses.
5. These non-adjacent couples are matched based on having closest proximity to each other.
6. Once the couple is identified, the in-law is distinguished. For child-child couples, assume the first listed spouse is the biological child to the household head, hence the second listed spouse is the child-in-law.

Table B2: Rules for Child-Parent Relationship Construction (after SPLOC is Generated)

Rule	Child's relationship to head	Parent's relationship to head	Age difference	Proximity requirement	Only applicable to 2002	Notes
Links involving Head, Spouse, and Grandparent (unambiguous)						
	Child	Head, spouse	No	No		
	Child-in-law	Head, spouse	No	No	Yes	4
	Head	Parent	No	No		
	Spouse	Parent	No	No		
	Sibling	Parent	No	No	Yes	
	Parent	Grandparent	No	No		
Links between grandchildren and children						
	Grandchild	Child, child-in-law	15–44	Weak		1
	Grandchild	Other relationship	15–44	Weak		1, 2
Links involving other relatives						
	Other relationship	Grandchild	15–44	Weak		1
	Other relationship	Other relative	15–44	Weak		1
	Other relationship	Sibling	15–44	Weak		1
	Other relationship	Child, child-in-law	15–44	Weak		1, 3

Notes:

1. Weak proximity requires that the child must be listed after its potential mothers (or potential fathers if it has no potential mothers). Among them, its mother is the one listed closest to the child.

2. Impose that no person with code "Child" is present in the household. The mother of the grandchildren in these cases tend to be listed as "Others" since there is no category for "Child-in-law."

3. Impose that no person with code "Grandchild" is present in the household. These cases tend to mix up the numerical code for "Grandchild" (code 6) and that for "Other relationship" (code 7).

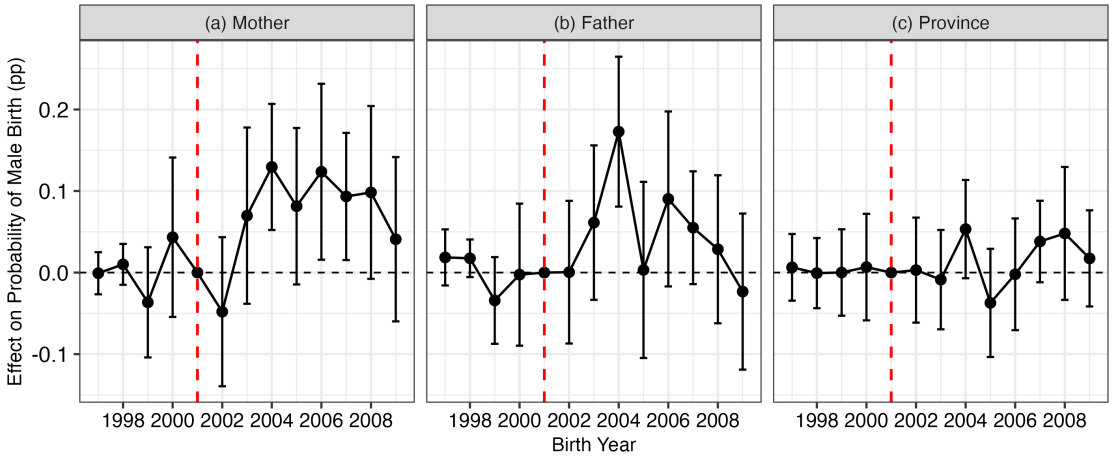
4. The head and spouse are the child-in-law's parents-in-law, not biological parents. This link identifies the household structure (the child-in-law married into the head's family) rather than a biological parent-child relationship. Children-in-law who have their own infants in the household are matched to those infants through the grandchild rules below.

Table B3: Performance of our algorithm in locating parents

	Freq.	Percent	Cum.
Both mom and dad correct	80021	88.95	88.95
Dad correct, mom incorrect	1831	2.04	90.99
Dad incorrect, mom correct	3853	4.28	95.27
Both mom and dad incorrect	4253	4.73	100.00
Total	89958	100.00	

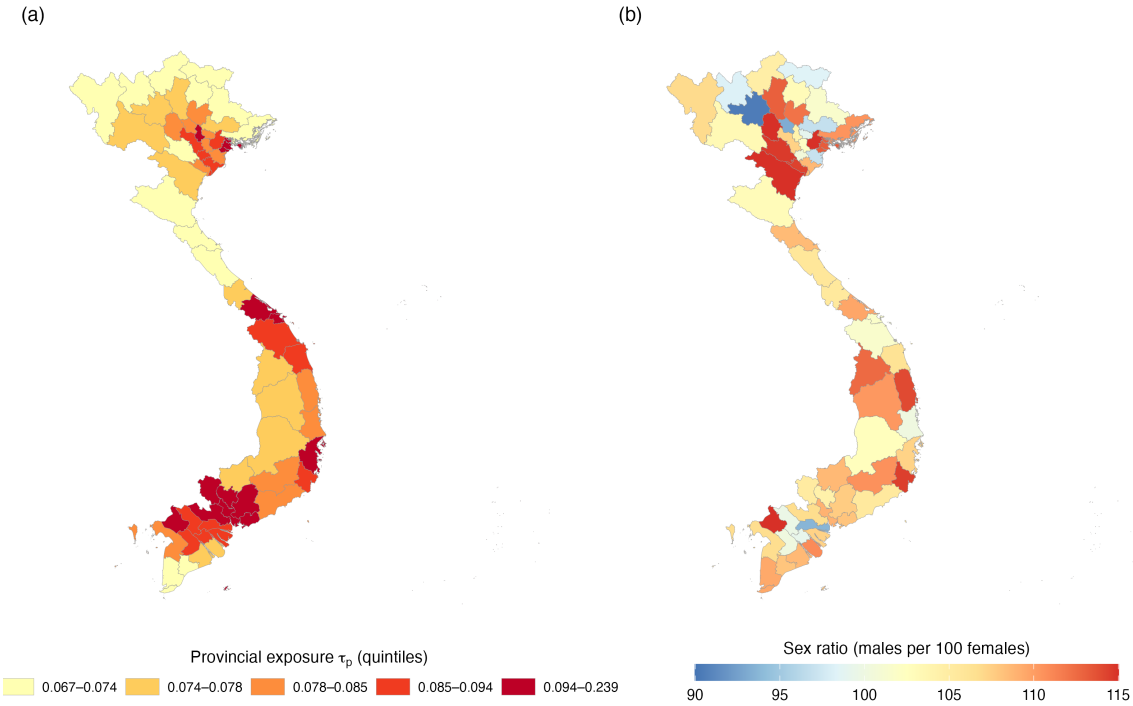
Notes: This table summarizes the comparison between the parent locators generated by our algorithm with the true parents locators provided by VHLSS 2014-2016 for children under 16 years old.

Figure C2: Individual Event Study Estimates of BTA Exposure Effects on Male Births



Notes: Separate event study regressions of MaleBirth on year dummies interacted with binary exposure for maternal (left), paternal (center), and provincial (right) tariff cut. Binary split at sample median. Province, birth-year, and parents' ISIC FE. Vertical dashed line marks BTA (late 2001). 95% CIs, province-clustered SE.

Figure C3: Provincial Exposure and Pre-BTA Sex Ratio at Birth



Notes: Panel (a) shows the provincial shift-share tariff exposure τ_p used to define P_p in the main specifications, displayed in quintiles. Panel (b) shows sex ratio at birth (males per 100 females) by province from 1997–1998 births in the 1999 Census. Provinces above the median exposure define the high-P_p group, and provinces above the median SRB define the high-pre-BTA-SRB subgroup in Figure 4.

Table C1: Comparison of Panel Sub-sample and Main Cross-section (2002 Baseline)

	Cross-section (2002 VHLSS)	Panel sub-sample (2002–2004–2006)
Mother's age	28.04	29.05
	(5.07)	(5.08)
Mother's education (yrs)	6.61	6.49
	(3.62)	(3.60)
Ethnic minority	0.230	0.233
	(0.421)	(0.423)
Urban	0.173	0.163
	(0.378)	(0.369)
Province exposure	0.09	0.09
	(0.02)	(0.02)
N	5,934	879

Notes: SD in parentheses. Cross-section uses 2002 VHLSS mothers of infants aged 0–1. Panel sub-sample is the 2002 baseline of the rotating panel, restricted to women aged 20–40 with stable residence and at least one birth before and after the BTA.

Table C2: Robustness to ISIC Industry Clustering

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$M_i \times$ Post	0.08***	0.15***	0.15***	-0.007***
	(0.03)	(0.04)	(0.03)	(0.002)
$F_i \times$ Post	0.03	-0.0003	0.02	0.005**
	(0.02)	(0.04)	(0.02)	(0.002)
$P_p \times$ Post	0.01	0.06**	-0.004	-0.002
	(0.01)	(0.02)	(0.01)	(0.0009)
R^2	0.02	0.25	0.60	0.29
Observations	63,894	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.7	0.25	0.02
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Binary treatment indicators (M_i, F_i, P_p) with controls. SE clustered by 2-digit ISIC industry only. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C3: Robustness to Province and ISIC Two-Way Clustering

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$M_i \times$ Post	0.08***	0.15***	0.15***	-0.007***
	(0.03)	(0.04)	(0.03)	(0.002)
$F_i \times$ Post	0.03	-0.0003	0.02	0.005**
	(0.02)	(0.03)	(0.02)	(0.002)
$P_p \times$ Post	0.01	0.06	-0.004	-0.002
	(0.01)	(0.04)	(0.02)	(0.001)
R^2	0.02	0.25	0.60	0.29
Observations	63,894	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.7	0.25	0.02
Mom FE				✓
Survey Year FE		✓	✓	
Birth Year FE	✓			✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Binary treatment indicators (M_i, F_i, P_p) with controls. SE two-way clustered by province and 2-digit ISIC. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C4: Robustness to Region-by-Year Fixed Effects

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$M_i \times$ Post	0.08**	0.15***	0.15***	-0.007***
	(0.03)	(0.05)	(0.03)	(0.001)
$F_i \times$ Post	0.04	0.04	0.002	0.005***
	(0.03)	(0.04)	(0.02)	(0.001)
$P_p \times$ Post	0.02	0.05	-0.002	-0.002
	(0.02)	(0.06)	(0.02)	(0.002)
R^2	0.02	0.26	0.61	0.29
Observations	63,894	17,472	17,665	2,011,963
Dep. Var. Mean	0.52	4.7	0.25	0.02
Mom FE				✓
Region-Year FE	✓	✓	✓	✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	
Controls	✓	✓	✓	✓

Notes: Binary treatment indicators (M_i, F_i, P_p) with region-by-year FE added to baseline. MaleBirth and labor outcomes include province and parents' ISIC FE. AnyBirth uses mother FE. Controls as in Table 3. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C5: Robustness to Logit Estimation for Male Birth Outcome

	MaleBirth	
	(1)	(2)
$M_i \times$ Post	0.22^{**}	0.33^{**}
	(0.10)	(0.13)
$F_i \times$ Post	0.11	0.13
	(0.12)	(0.13)
$P_p \times$ Post	0.006	0.06
	(0.05)	(0.07)
Observations	66,554	63,886
Dep. Var. Mean	0.52	0.52
Birth Year FE	✓	✓
Dad ISIC FE	✓	✓
Mom ISIC FE	✓	✓
Province FE	✓	✓
Controls		✓

Notes: Logit estimates with binary treatment indicators (M_i, F_i, P_p). Birth-year, province, and parents' ISIC FE. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C6: Robustness to Continuous Tariff Exposure Measures

	MaleBirth					
	(1)	(2)	(3)	(4)	(5)	(6)
$\tau_m \times$ Post	0.216**		0.205**	0.385***	0.212**	0.384***
	(0.093)		(0.090)	(0.134)	(0.086)	(0.137)
$\tau_d \times$ Post		0.100	0.049	0.093	0.046	0.091
		(0.127)	(0.126)	(0.133)	(0.127)	(0.133)
$\tau_p \times$ Post					-0.003	0.002
					(0.008)	(0.012)
R^2	0.015	0.014	0.015	0.016	0.015	0.016
Observations	66,562	66,562	66,562	63,894	66,562	63,894
Dep. Var. Mean	0.518	0.518	0.518	0.518	0.518	0.518
Controls				✓		✓

Notes: Dep. var. is MaleBirth. Continuous treatments (τ_m, τ_d, τ_p) measure tariff reduction exposure. Columns progressively add treatments and controls as in Table 3. Same FE as baseline. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C7: Robustness to Continuous Tariff Exposure on Maternal Labor Market Outcomes

	MomWageEmp			log(MomWorkHrs)		
	(1)	(2)	(3)	(4)	(5)	(6)
$\tau_m \times$ Post	0.509***	0.629***	0.630***	0.146	0.325	0.349
	(0.146)	(0.154)	(0.153)	(0.150)	(0.242)	(0.238)
$\tau_d \times$ Post		-0.070	-0.070		0.143	0.150
		(0.075)	(0.075)		(0.180)	(0.180)
$\tau_p \times$ Post			-0.002			-0.082**
			(0.023)			(0.039)
R^2	0.600	0.603	0.603	0.246	0.253	0.253
Observations	17,665	17,665	17,665	17,472	17,472	17,472
Dep. Var. Mean	0.248	0.248	0.248	4.66	4.66	4.66
Controls		✓	✓		✓	✓

Notes: Dep. vars. are MomWageEmp (cols. 1–3) and log monthly hours (cols. 4–6). Continuous treatments (τ_m, τ_d, τ_p). Columns progressively add treatments and controls. Survey-year, province, and parents' ISIC FE. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C8: Robustness to Continuous Tariff Exposure on Fertility

	AnyBirth					
	(1)	(2)	(3)	(4)	(5)	(6)
$\tau_m \times$ Post	-0.003		-0.008**	-0.011**	-0.009***	-0.012**
	(0.003)		(0.004)	(0.005)	(0.003)	(0.005)
$\tau_d \times$ Post		0.021***	0.023***	0.026***	0.023***	0.026***
		(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\tau_p \times$ Post					0.0009***	0.004***
					(0.0003)	(0.001)
R^2	0.286	0.286	0.286	0.287	0.286	0.287
Observations	2,011,978	2,011,992	2,011,963	2,011,963	2,011,963	2,011,963
Dep. Var. Mean	0.018	0.018	0.018	0.018	0.018	0.018
Controls				✓		✓

Notes: Dep. var. is AnyBirth. Continuous treatments (τ_m, τ_d, τ_p). Columns progressively add treatments as in Table C6. Mother and birth-year FE. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C9: Fertility Results with Mother-Level Clustering (Binary Treatment)

	AnyBirth					
	(1)	(2)	(3)	(4)	(5)	(6)
$M_i \times$ Post	-0.004***		-0.006***	-0.007***	-0.006***	-0.007***
	(0.0008)		(0.0009)	(0.0009)	(0.0009)	(0.0009)
$F_i \times$ Post		0.003***	0.005***	0.005***	0.005***	0.005***
		(0.0010)	(0.001)	(0.001)	(0.001)	(0.001)
$P_p \times$ Post					-0.0001	-0.002**
					(0.0006)	(0.0007)
R^2	0.29	0.29	0.29	0.29	0.29	0.29
Observations	2,011,963	2,011,963	2,011,963	2,011,963	2,011,963	2,011,963
Dep. Var. Mean	0.02	0.02	0.02	0.02	0.02	0.02
Controls				✓		✓

Notes: Same as Table 5 with SE clustered at the mother level. Mother and birth-year FE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C10: Relative-Returns Test for All Outcomes with All Four Exposure Channels

	MaleBirth	log(MomWorkHrs)	MomWageEmp	AnyBirth
	(1)	(2)	(3)	(4)
$M_i \times$ Post	0.08**	0.16***	0.15***	-0.007***
	(0.03)	(0.05)	(0.03)	(0.001)
$F_i \times$ Post	0.03	0.004	0.01	0.005***
	(0.03)	(0.04)	(0.02)	(0.001)
$P_p \times$ Post	0.009	0.11*	-0.04	-0.002
	(0.01)	(0.07)	(0.03)	(0.002)
$P_p^w \times$ Post	0.009	-0.09	0.05**	0.0008
	(0.01)	(0.07)	(0.03)	(0.002)
R^2	0.02	0.25	0.60	0.29
Observations	63,894	17,198	17,389	2,011,963
Dep. Var. Mean	0.52	4.7	0.25	0.02
Mom FE				✓
Dad ISIC FE	✓	✓	✓	
Mom ISIC FE	✓	✓	✓	
Province FE	✓	✓	✓	

Notes: One controlled specification per outcome. M_i, F_i, P_p = binary above-median maternal, paternal, and provincial BTA exposure. P_p^w = above-median female-weighted provincial exposure ($\sum_j f_j \cdot s_{jp} \cdot \tau_j$, f_j = pre-BTA female employment share). Post = birth year > 2001 (MaleBirth, AnyBirth) or survey year > 2002 (labor). Cols. 1–3 include province, parents' ISIC, and time FE. Col. 4 includes mother and birth-year FE. Controls as in Table 3. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C11: Sensitivity of Fertility Estimates to Retrospective Lookback Window

	AnyBirth				
	$K \leq 10$	$K \leq 12$	$K \leq 15$	$K \leq 20$	Full
	(1)	(2)	(3)	(4)	(5)
$M_i \times$ Post	-0.003*	-0.005***	-0.006***	-0.007***	-0.007***
	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)
$F_i \times$ Post	0.006***	0.005***	0.005***	0.005***	0.005***
	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)
$P_p \times$ Post	-0.004*	-0.003	-0.002	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
R^2	0.307	0.297	0.290	0.287	0.287
Observations	1,446,466	1,632,298	1,844,307	2,008,581	2,011,963
Dependent variable mean	0.018	0.018	0.018	0.018	0.018
Birth Year FE	✓	✓	✓	✓	✓
Mother FE	✓	✓	✓	✓	✓

Notes: Each column restricts the retrospective fertility panel to woman-years where the implied child age at survey is at most K years. Column (5) is the unrestricted full sample. All specifications include mother and birth-year fixed effects, world-average tariff controls, and province-clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table C12: Pre-Treatment Bargaining Power and Sex Selection, 1999 Census

	Mom $\leq$ Dad education	Mom $>$ Dad education
Proportion male	0.5125	0.5145
	(0.0029)	(0.0049)
N	30,548	10,619
Regression. Dep. var. = Male birth		
Mom $>$ Dad education	0.0026	
	(0.0048)	
N	41,167	
Province FE	Yes	

Notes: 1997–1998 births from the 1999 Vietnam Census (IPUMS). "Mom $>$ Dad education" = 1 if mother's education exceeds father's. Top panel reports group means (SE of mean in parentheses). Bottom panel reports OLS with province FE. Province-clustered SE. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.